

BFS Traversal

Step 1: Start at E

Legend

Complete

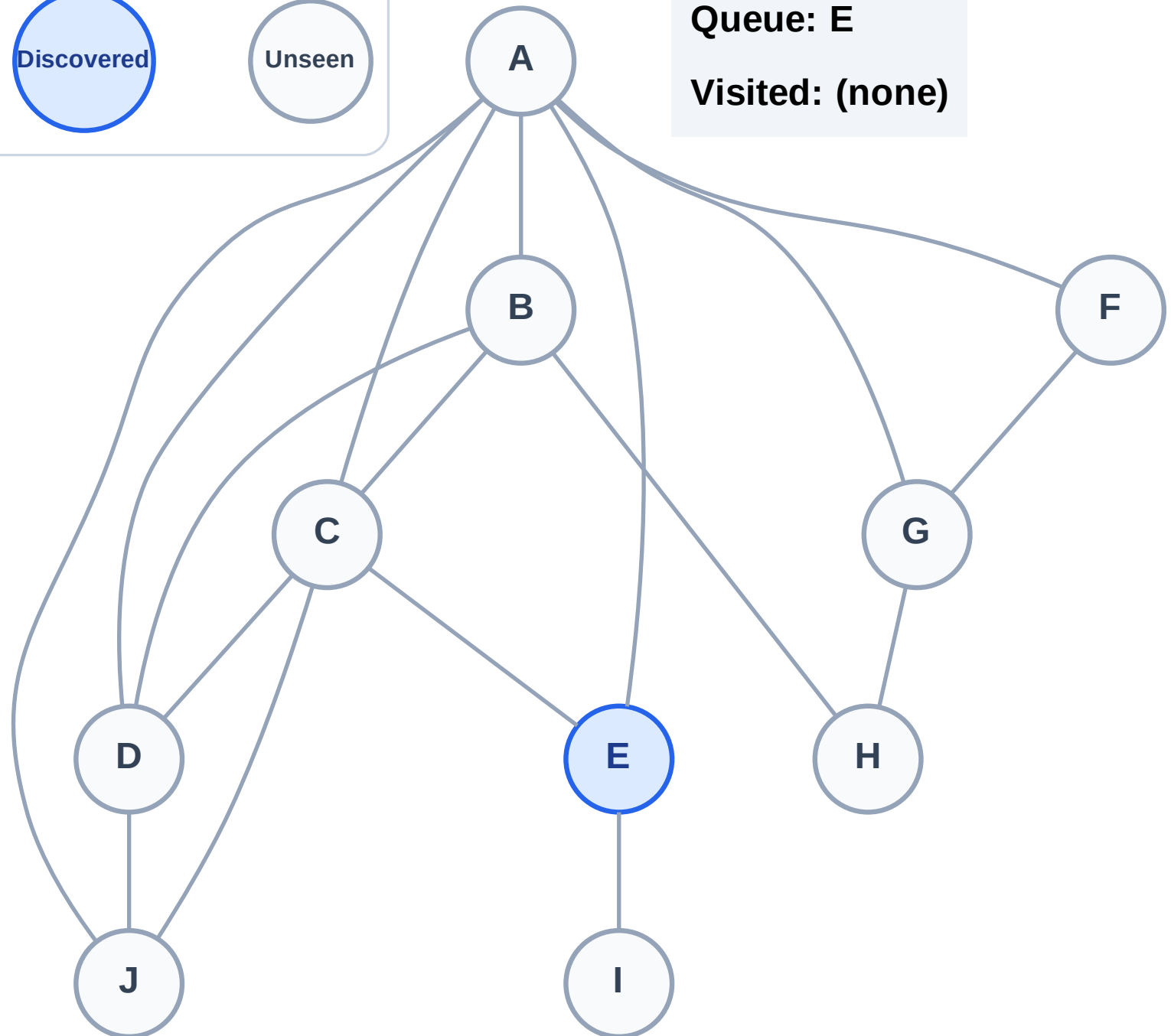
Processing

Discovered

Unseen

Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

Complete

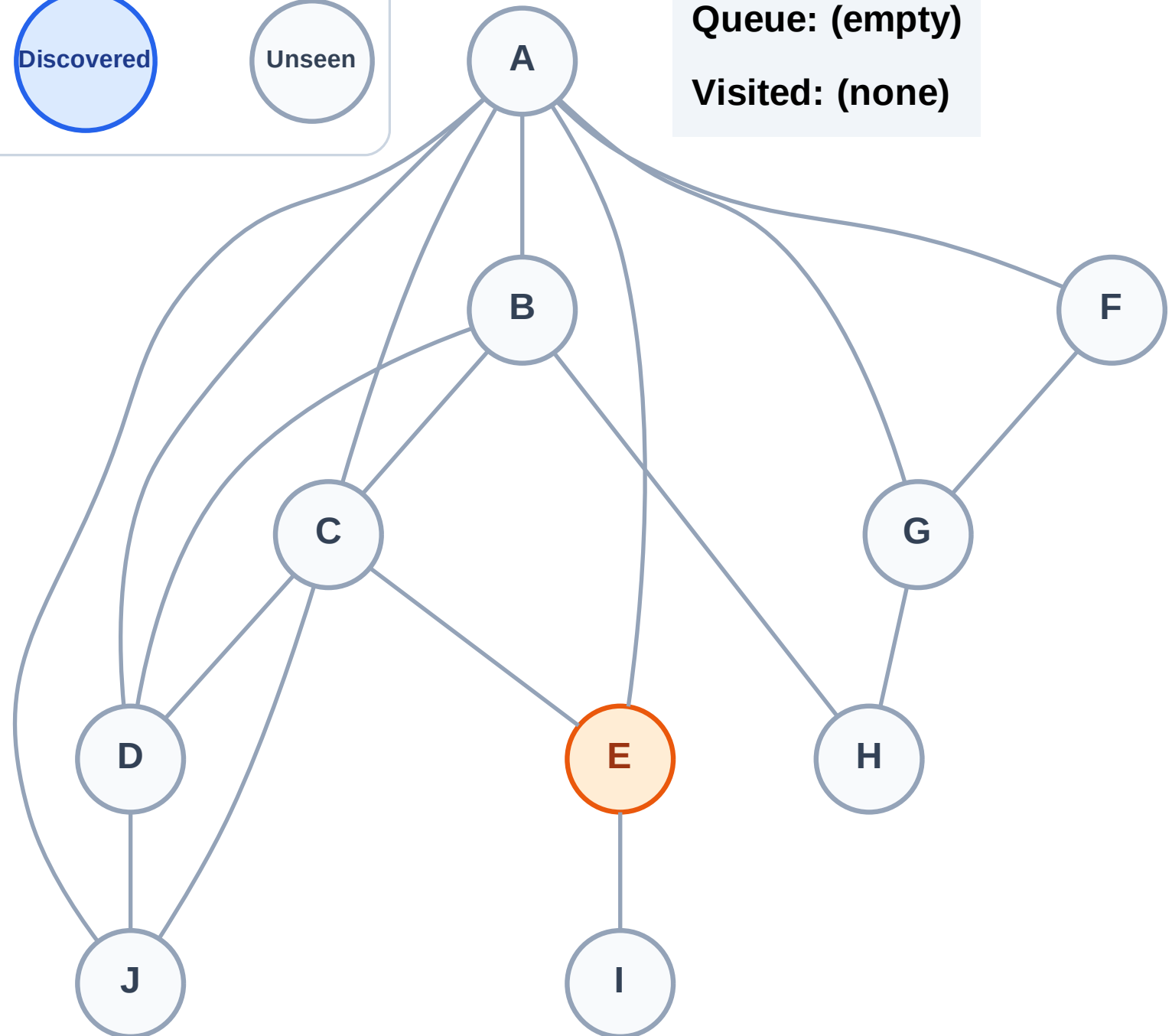
Processing

Discovered

Unseen

Queue: (empty)

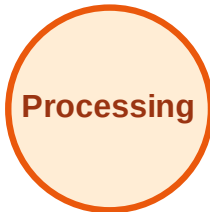
Visited: (none)



BFS Traversal

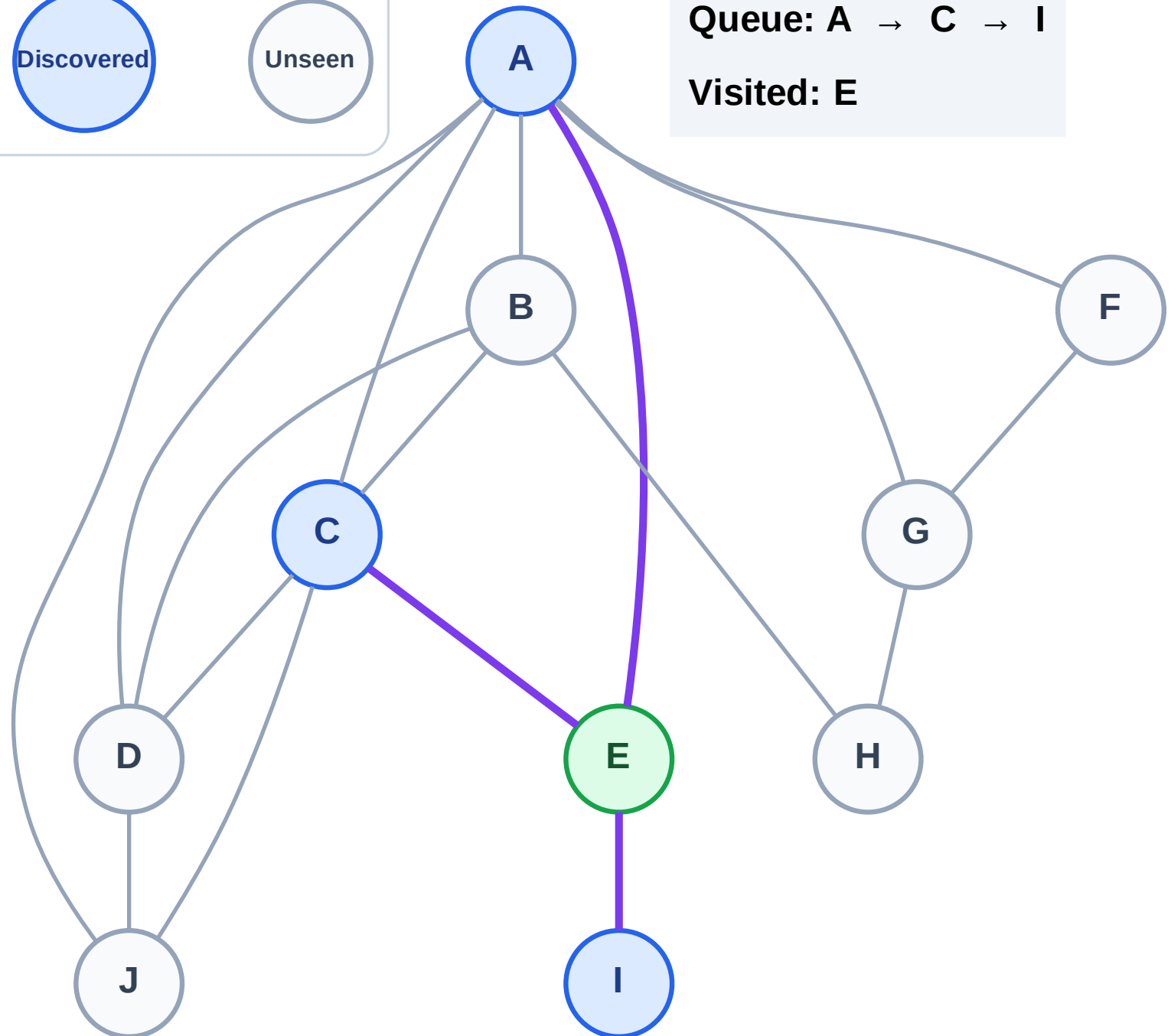
Step 3: Discover A, C, I from E

Legend



Queue: A → C → I

Visited: E



BFS Traversal

Step 4: Process node A

Legend

Complete

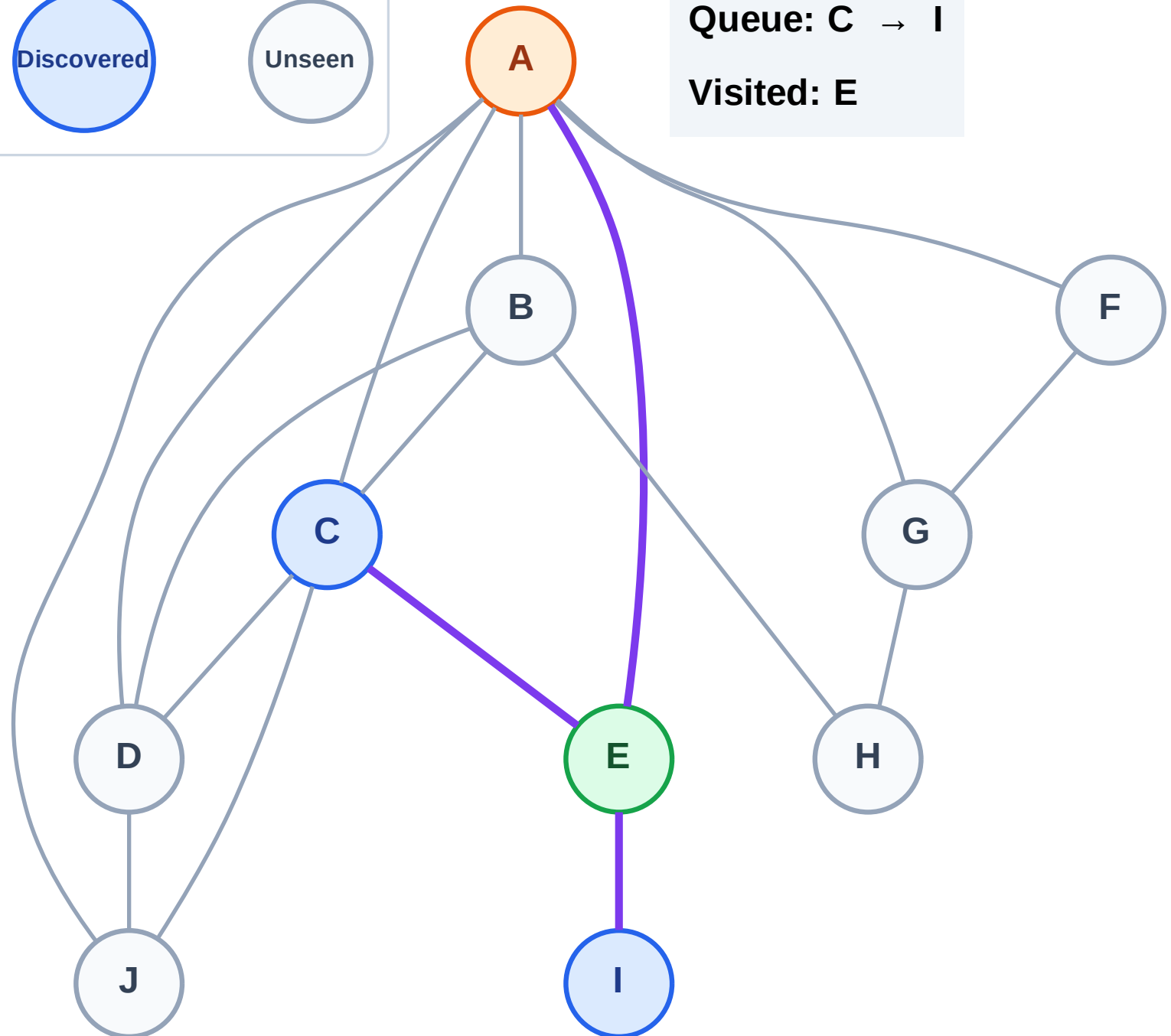
Processing

Discovered

Unseen

Queue: C → I

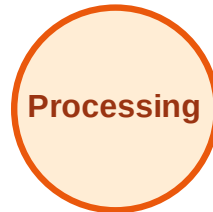
Visited: E



BFS Traversal

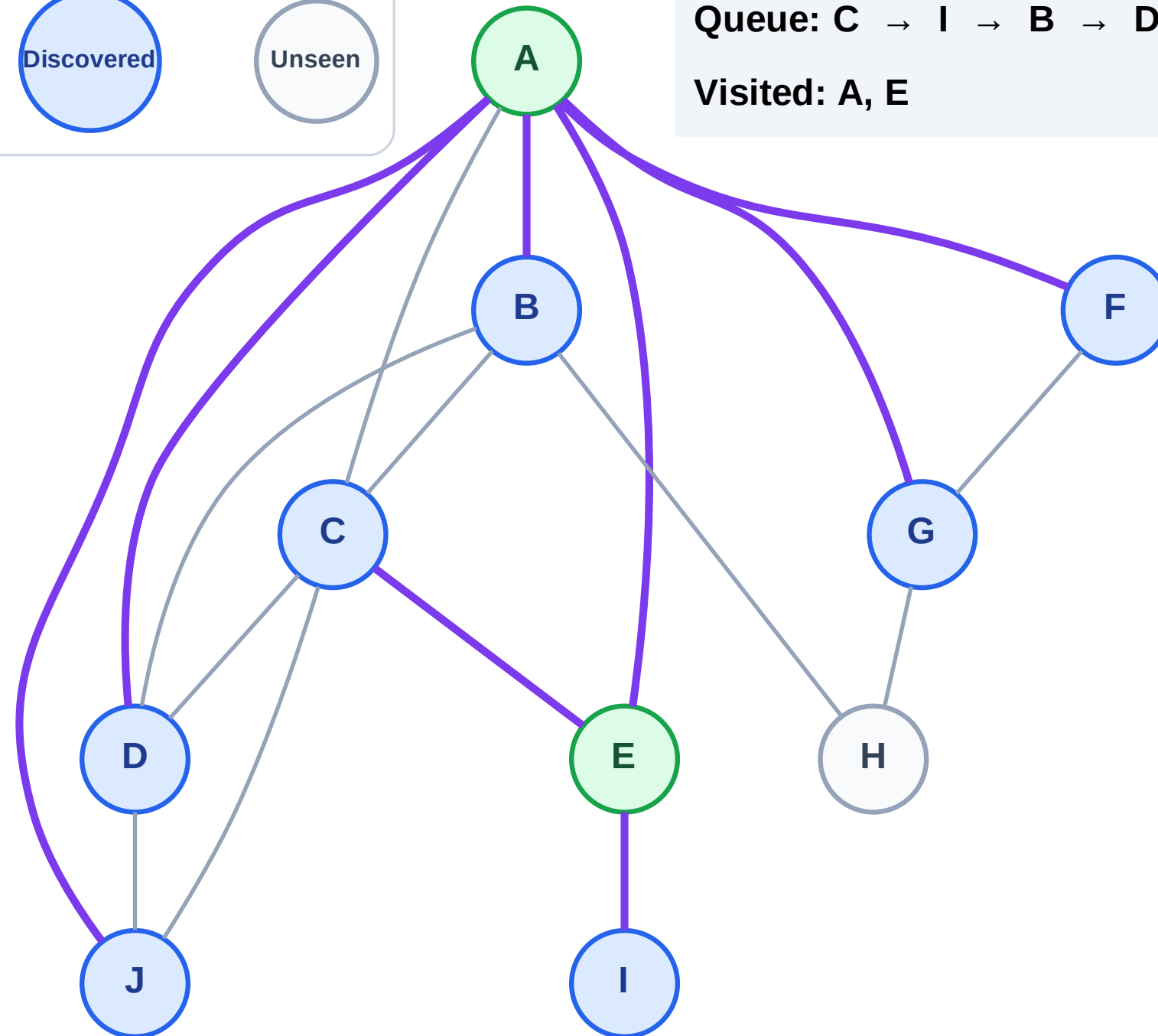
Step 5: Discover B, D, F, G, J from A

Legend



Queue: C → I → B → D → F → G → J

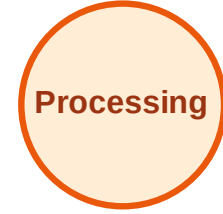
Visited: A, E



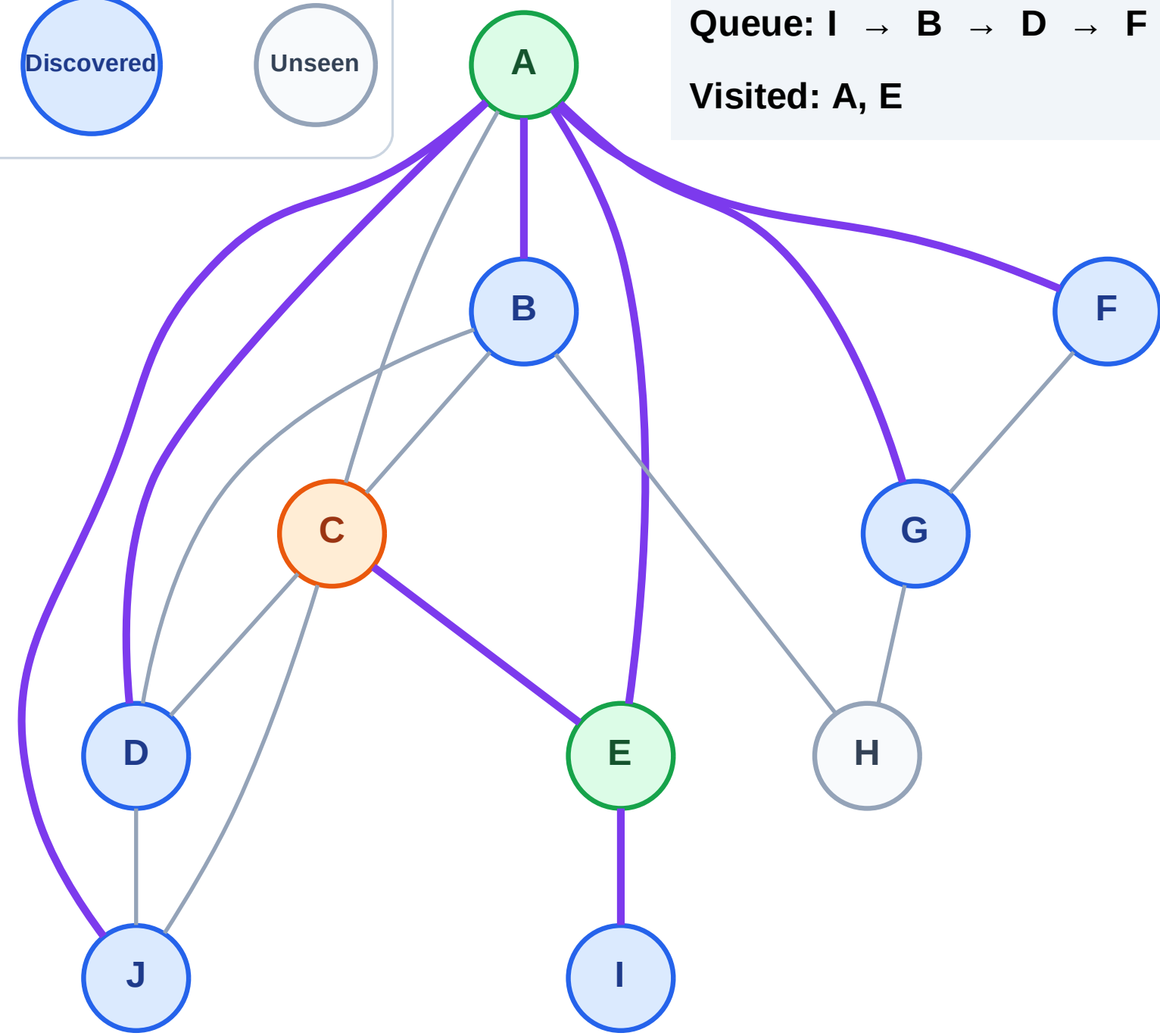
BFS Traversal

Step 6: Process node C

Legend



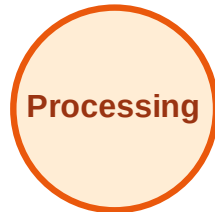
Queue: I → B → D → F → G → J
Visited: A, E



BFS Traversal

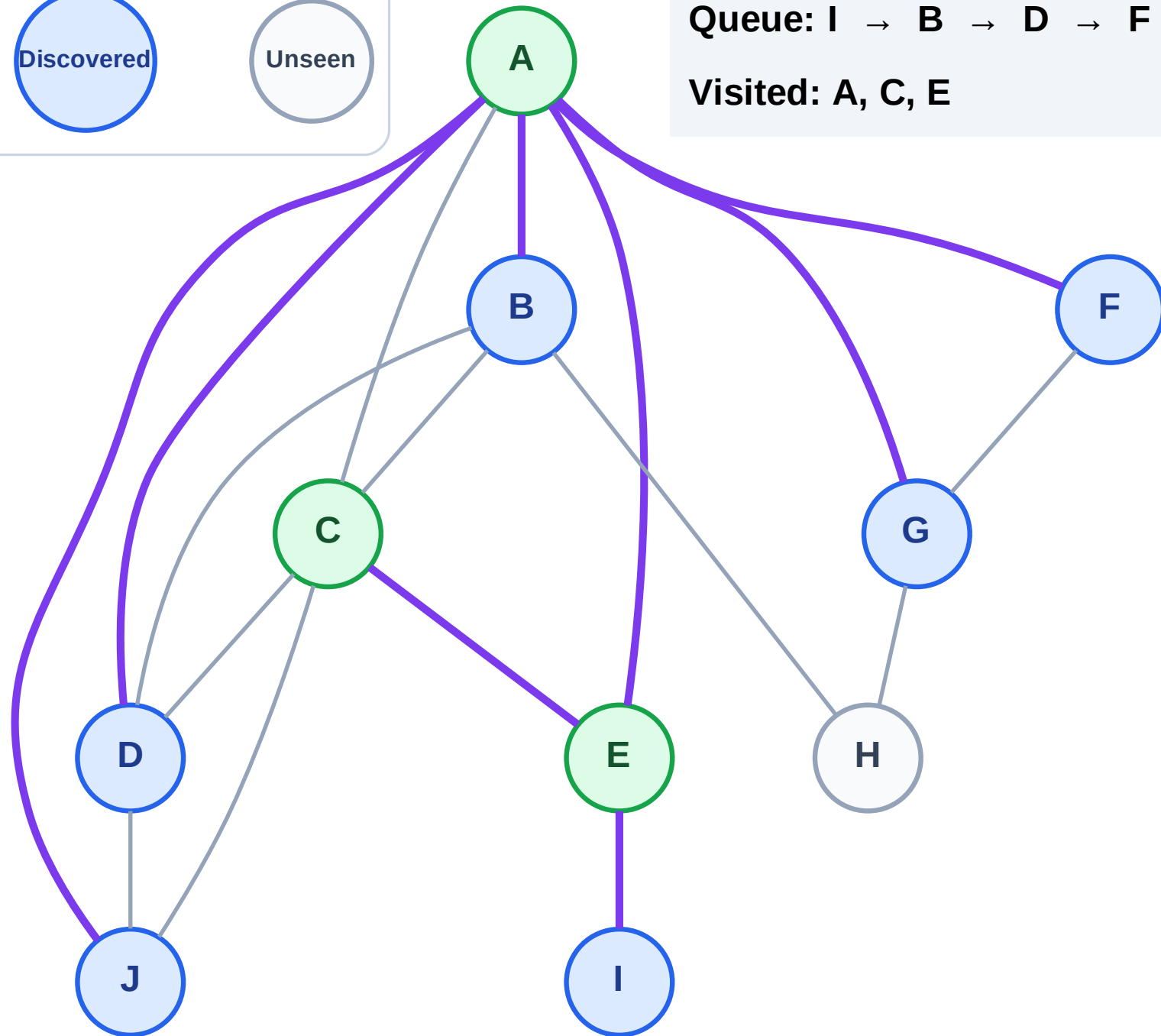
Step 7: No new neighbors from C

Legend



Queue: I → B → D → F → G → J

Visited: A, C, E



BFS Traversal

Step 8: Process node I

Legend

Complete

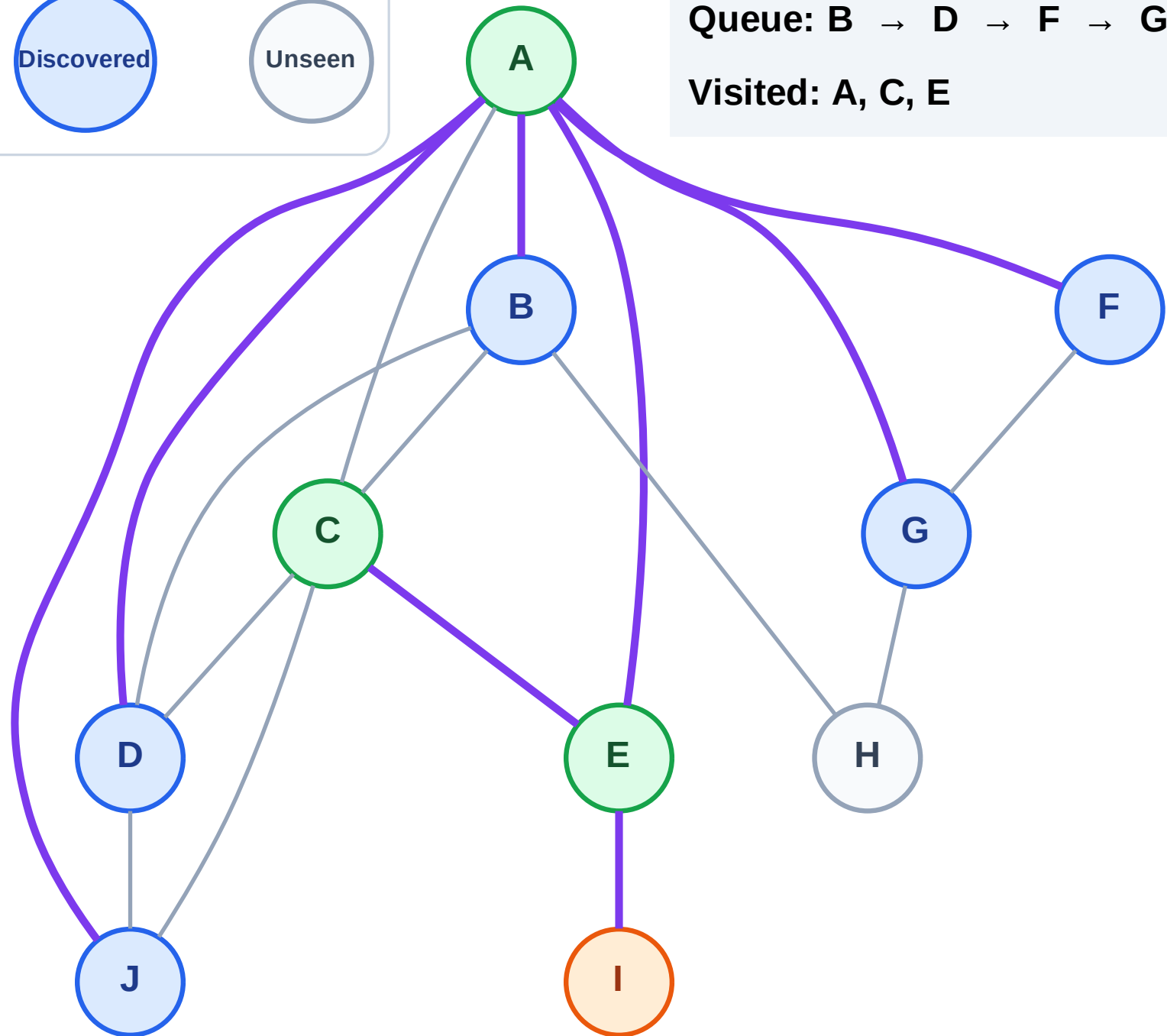
Processing

Discovered

Unseen

Queue: B → D → F → G → J

Visited: A, C, E



BFS Traversal

Step 9: No new neighbors from I

Legend

Complete

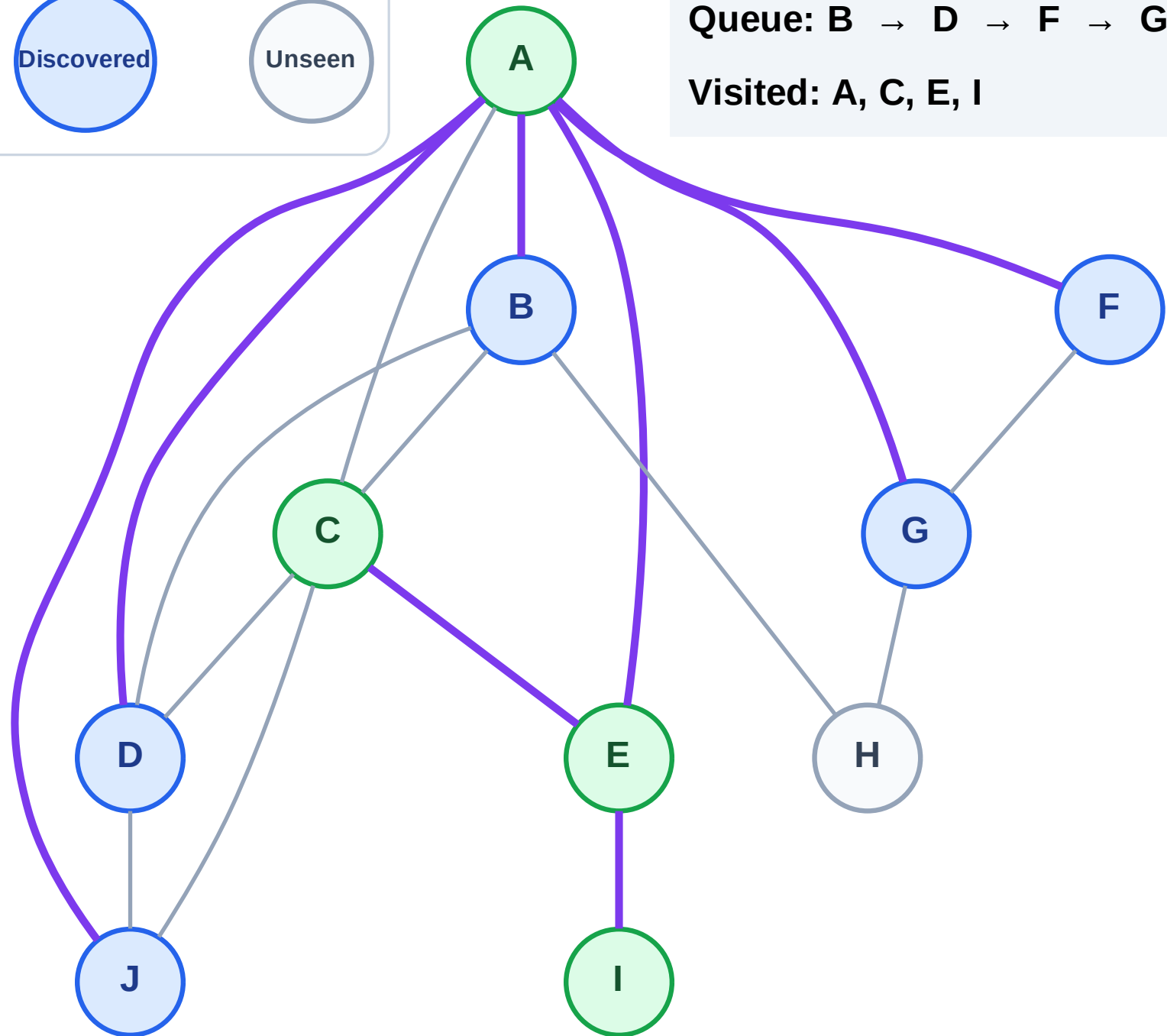
Processing

Discovered

Unseen

Queue: B → D → F → G → J

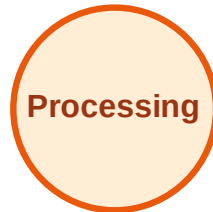
Visited: A, C, E, I



BFS Traversal

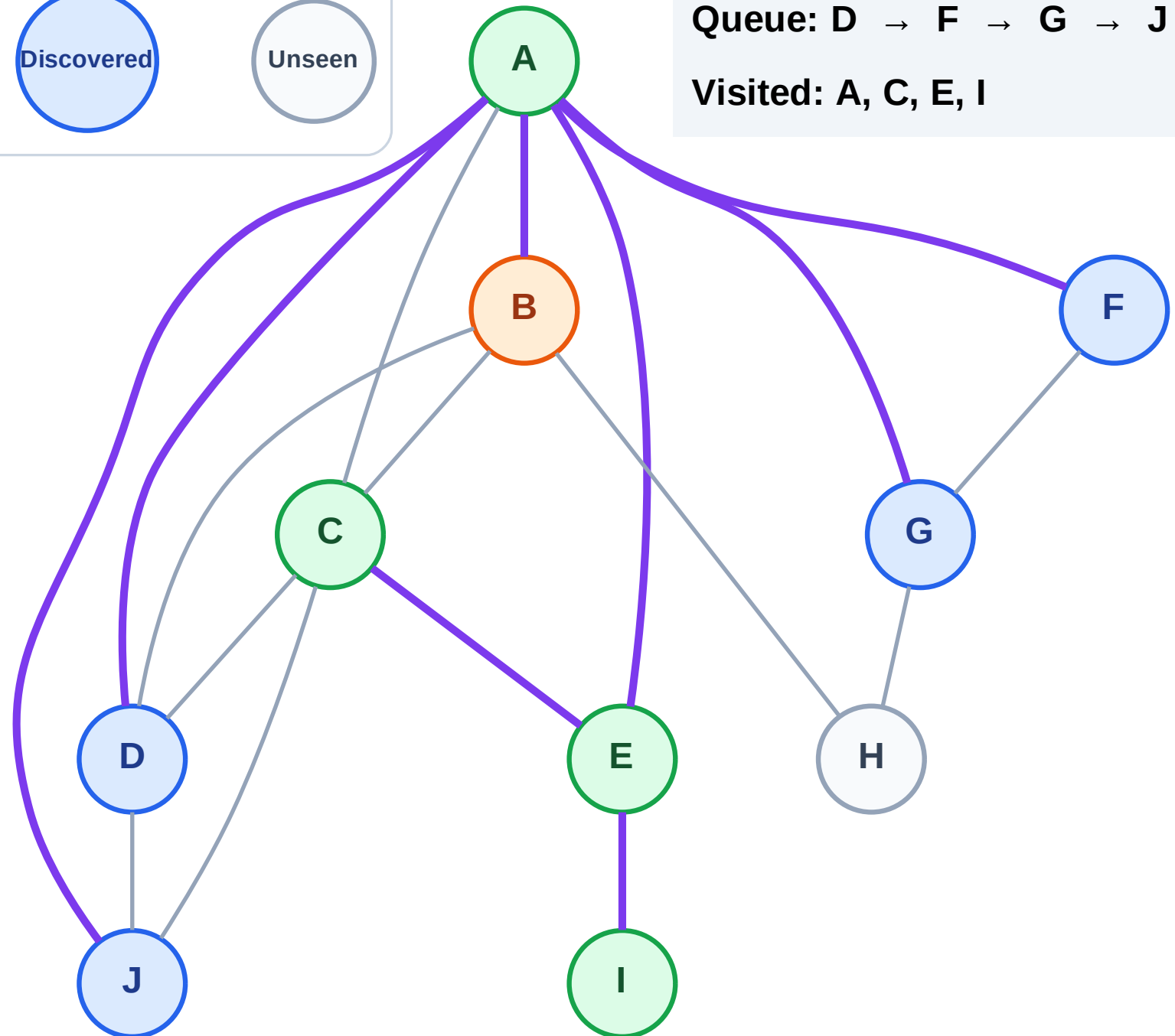
Step 10: Process node B

Legend



Queue: D → F → G → J

Visited: A, C, E, I



BFS Traversal

Step 11: Discover H from B

Legend

Complete

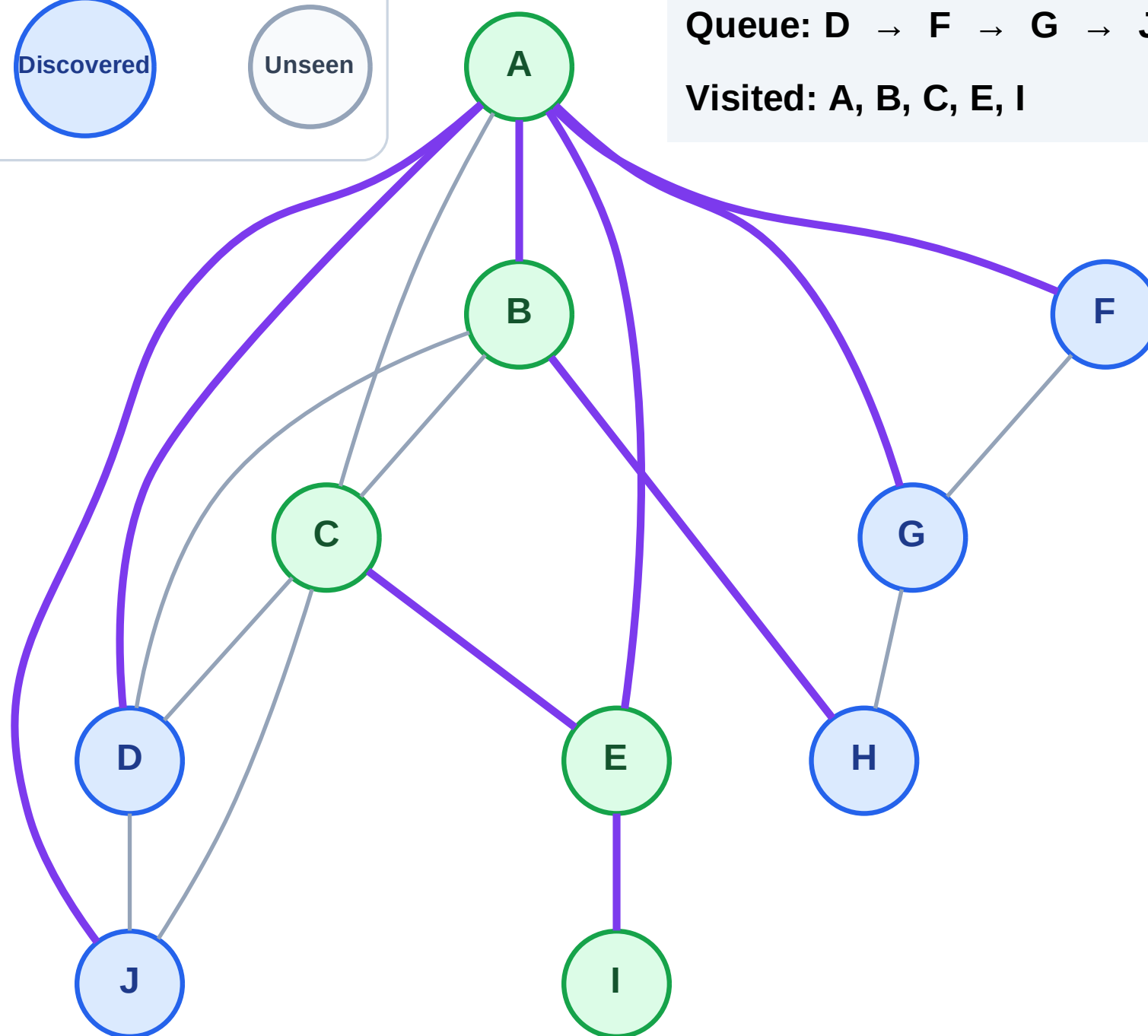
Processing

Discovered

Unseen

Queue: D → F → G → J → H

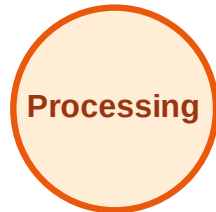
Visited: A, B, C, E, I



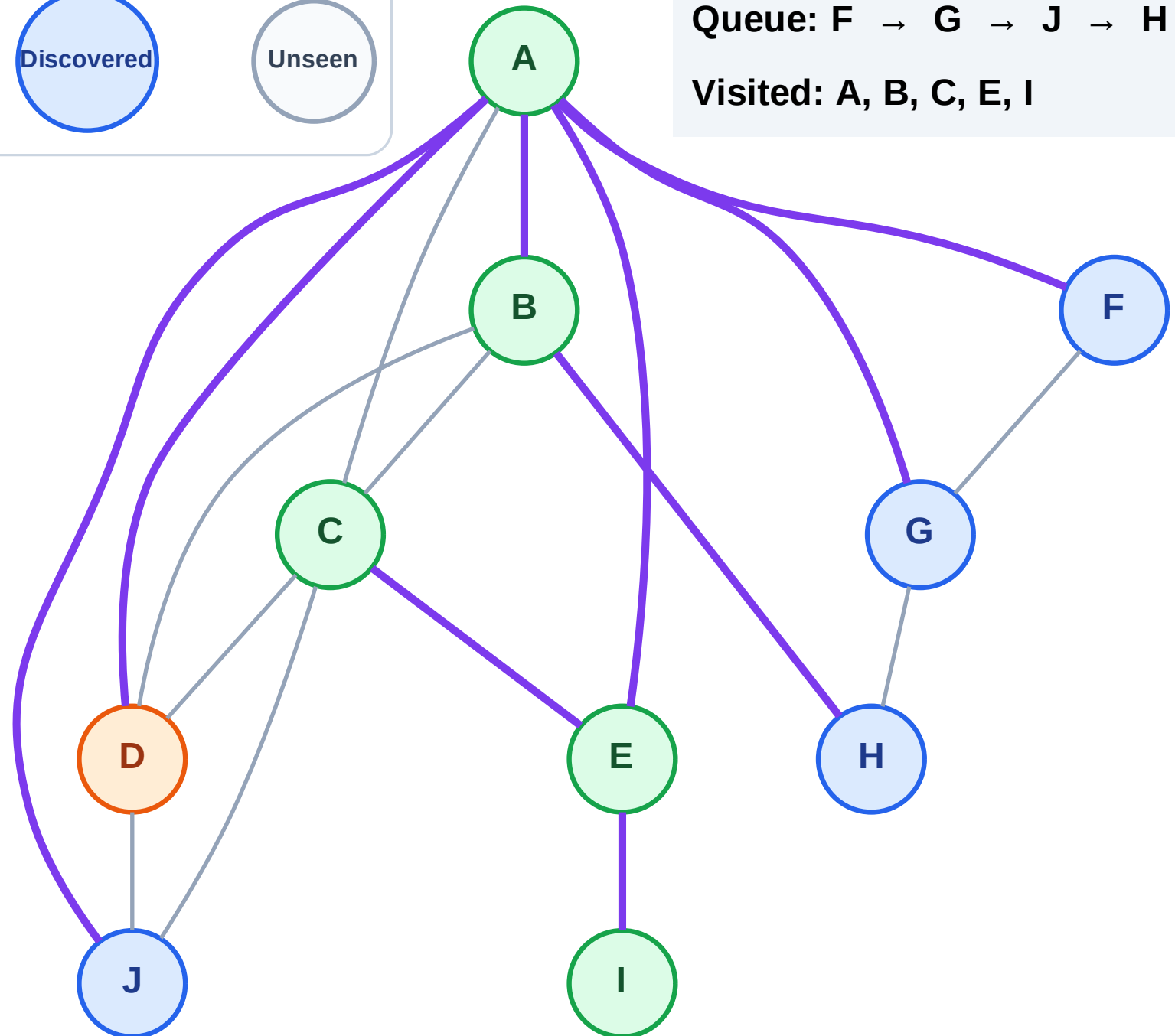
BFS Traversal

Step 12: Process node D

Legend



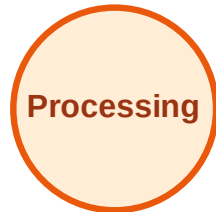
Queue: F → G → J → H
Visited: A, B, C, E, I



BFS Traversal

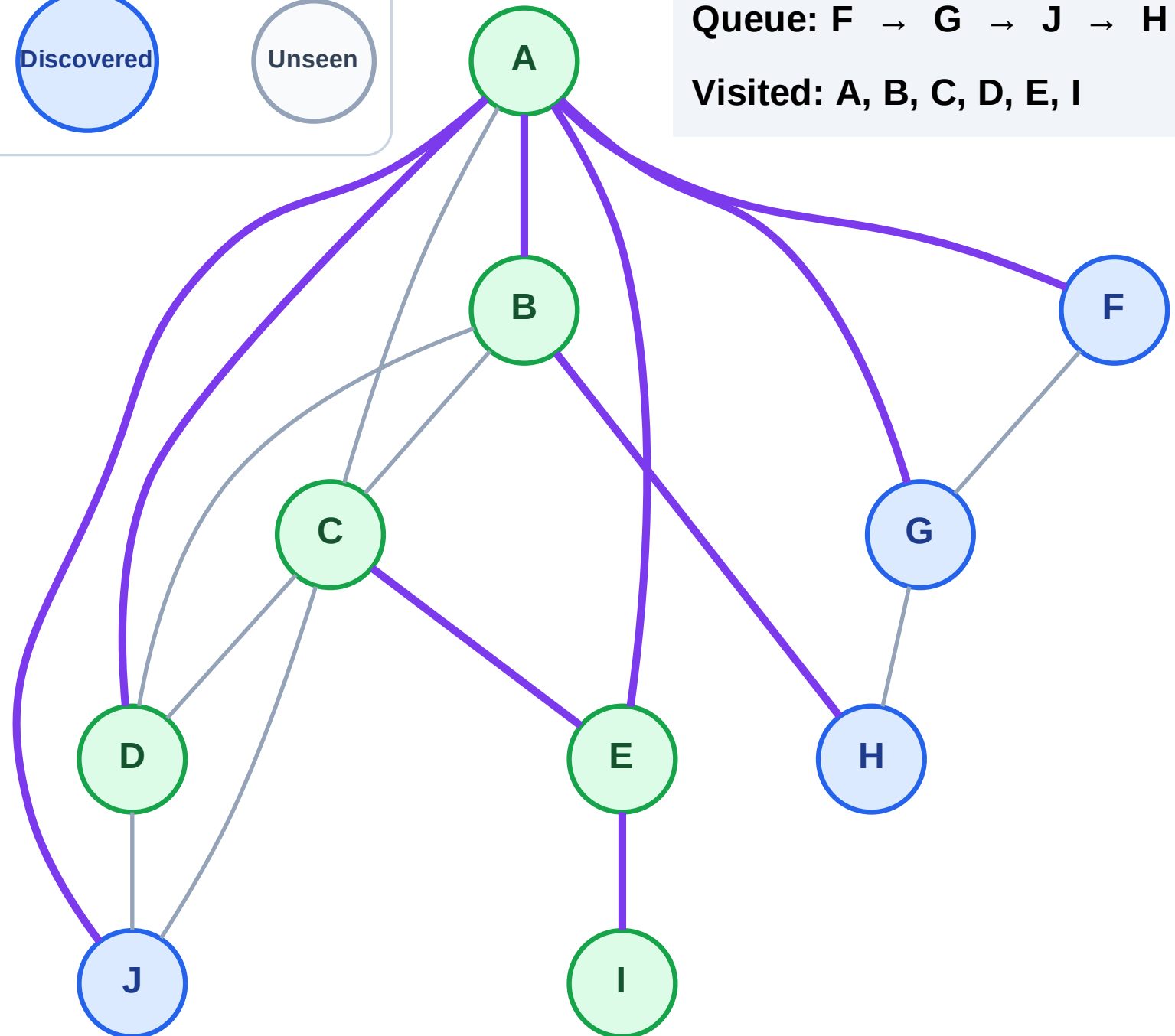
Step 13: No new neighbors from D

Legend



Queue: F → G → J → H

Visited: A, B, C, D, E, I



BFS Traversal

Step 14: Process node F

Legend

Complete

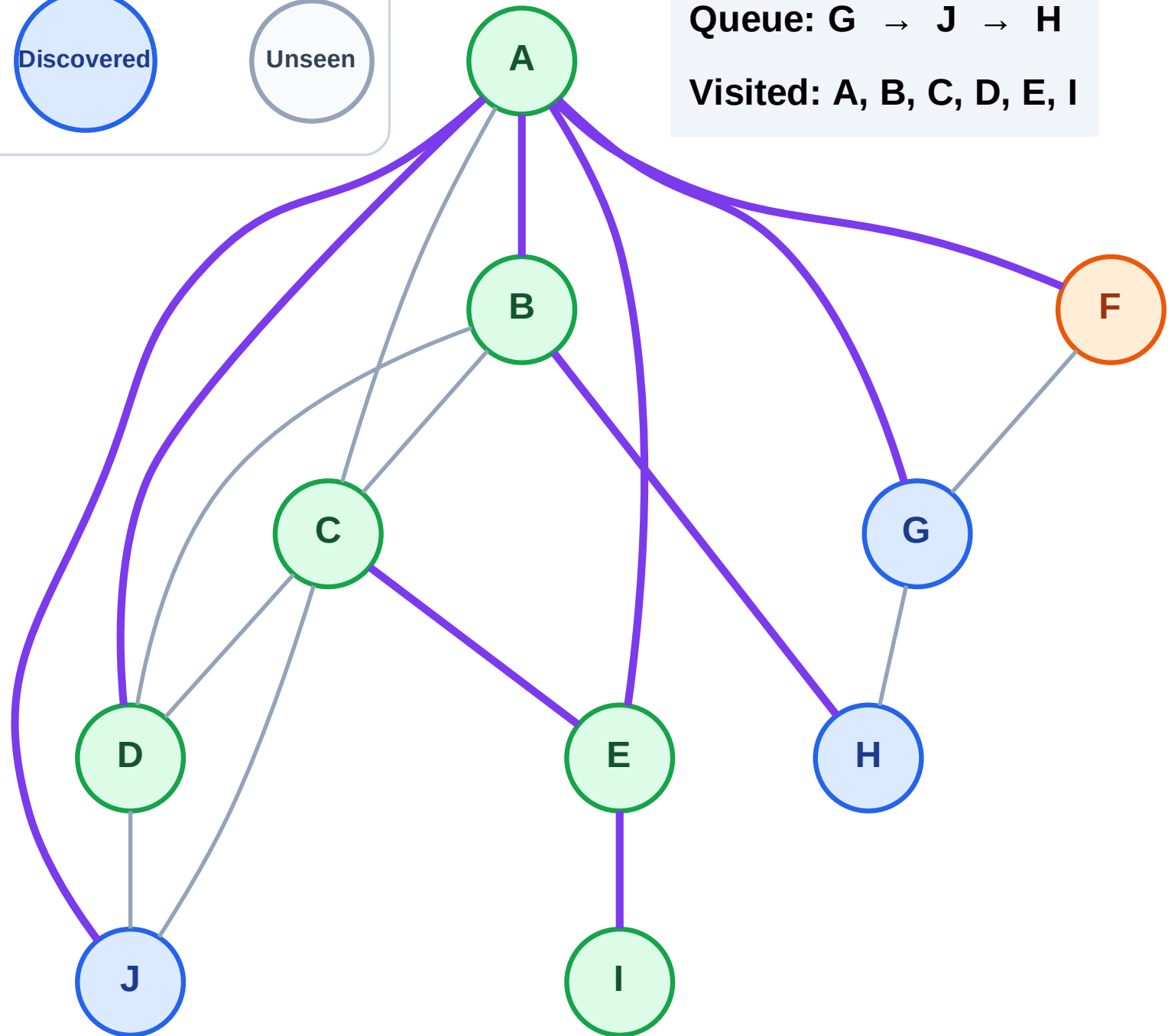
Processing

Discovered

Unseen

Queue: G → J → H

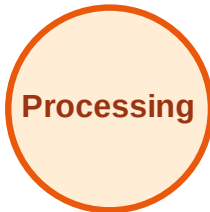
Visited: A, B, C, D, E, I



BFS Traversal

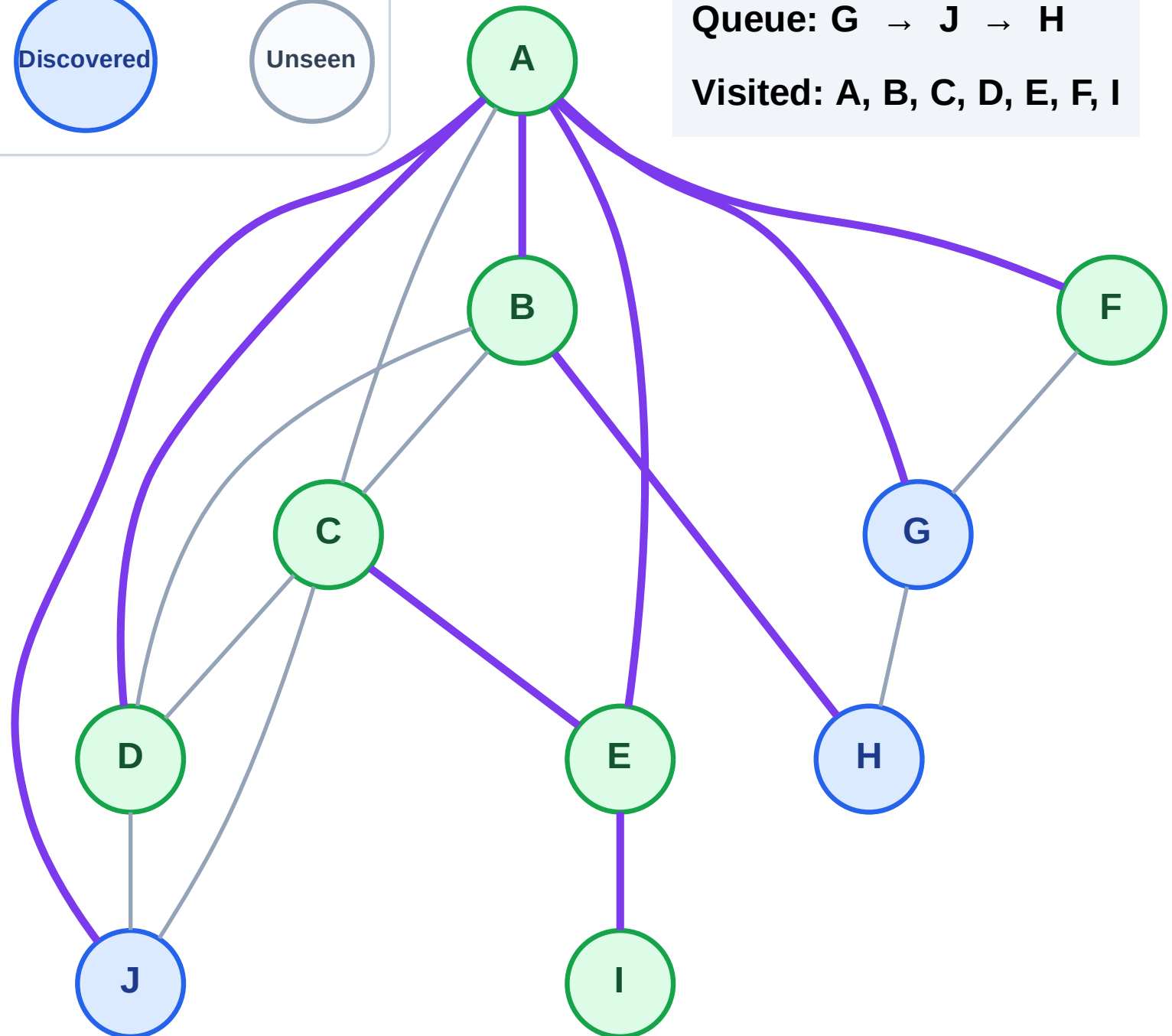
Step 15: No new neighbors from F

Legend



Queue: G → J → H

Visited: A, B, C, D, E, F, I



BFS Traversal

Step 16: Process node G

Legend

Complete

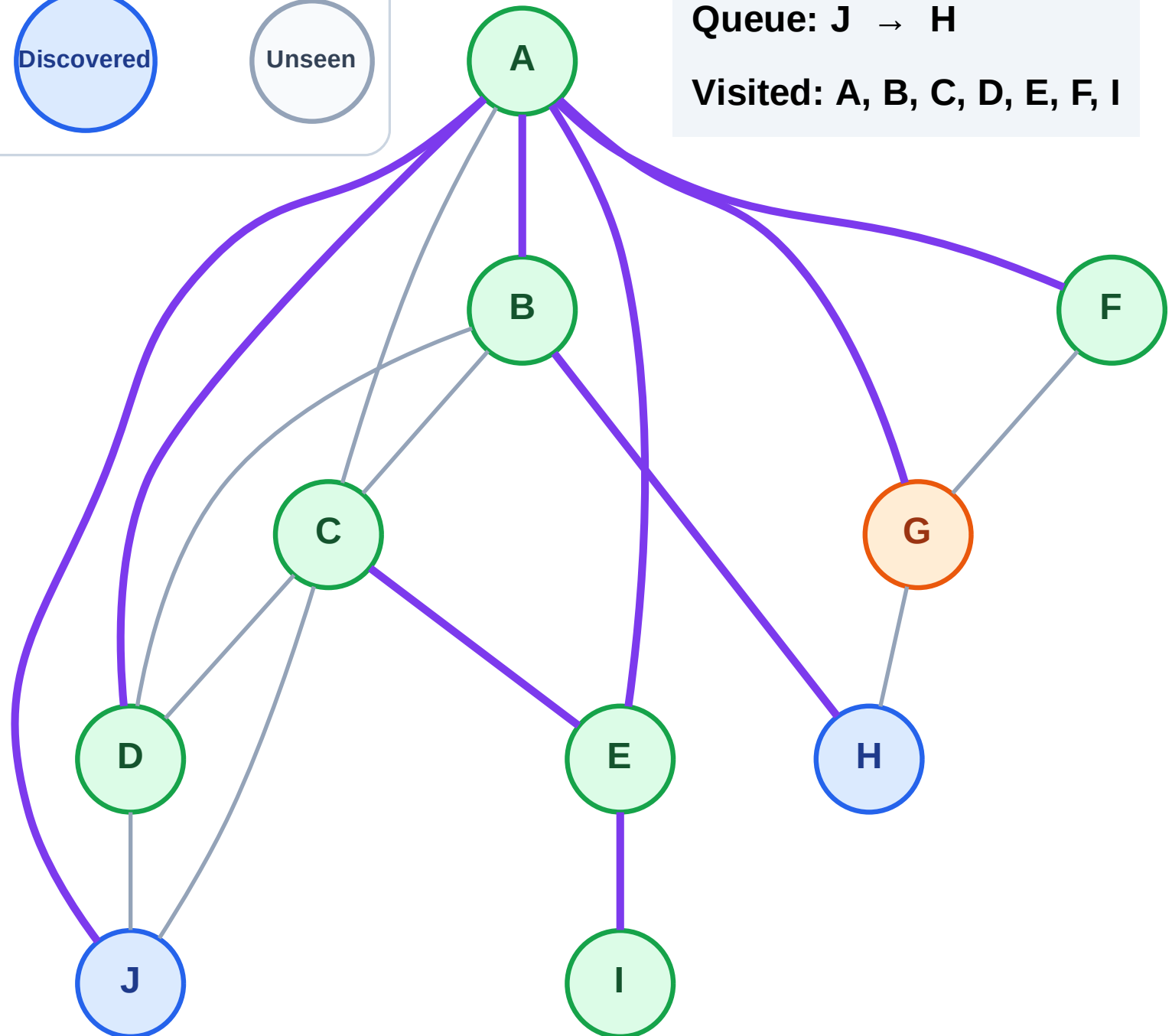
Processing

Discovered

Unseen

Queue: J → H

Visited: A, B, C, D, E, F, I



BFS Traversal

Step 17: No new neighbors from G

Legend

Complete

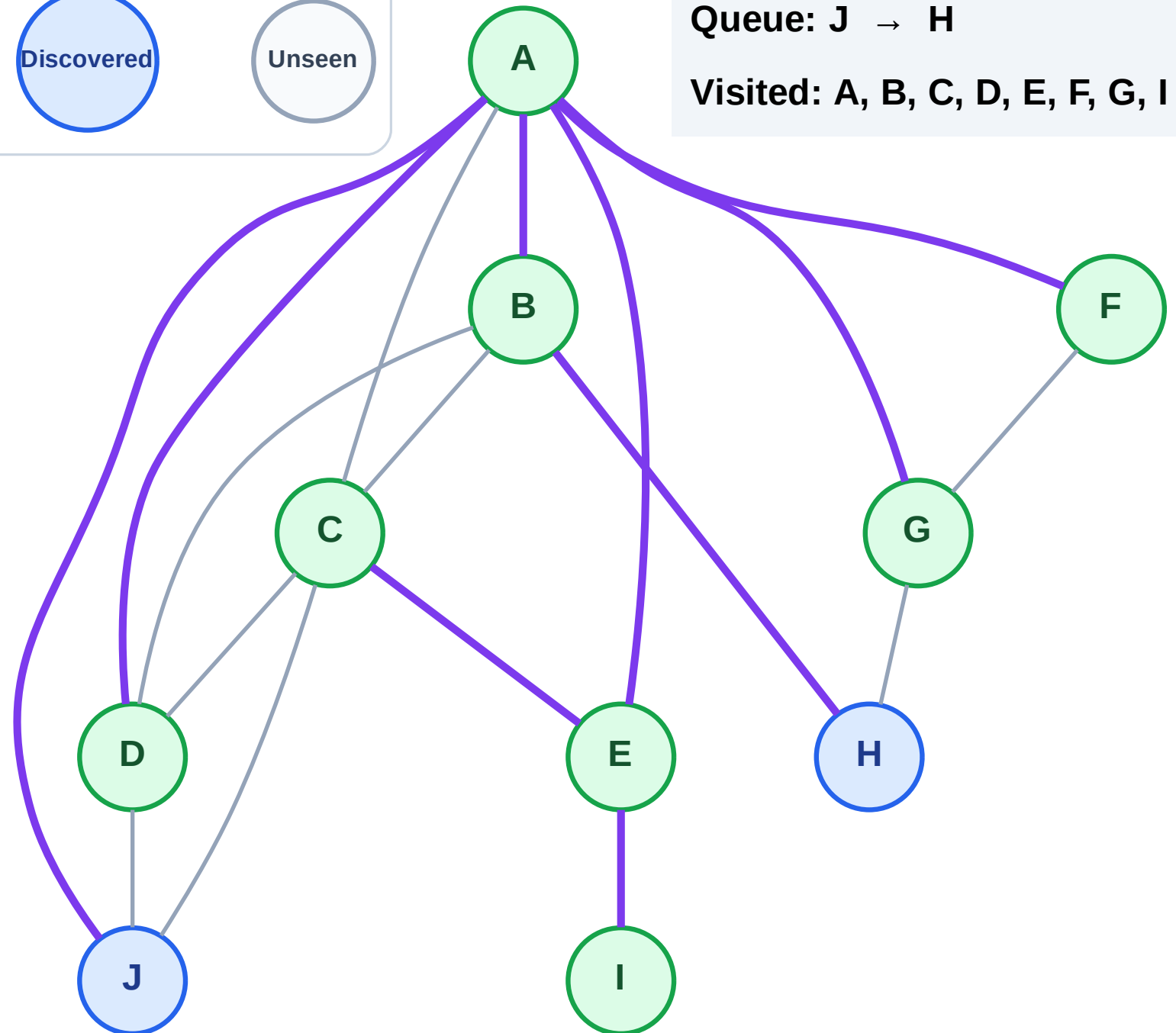
Processing

Discovered

Unseen

Queue: J → H

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 18: Process node J

Legend

Complete

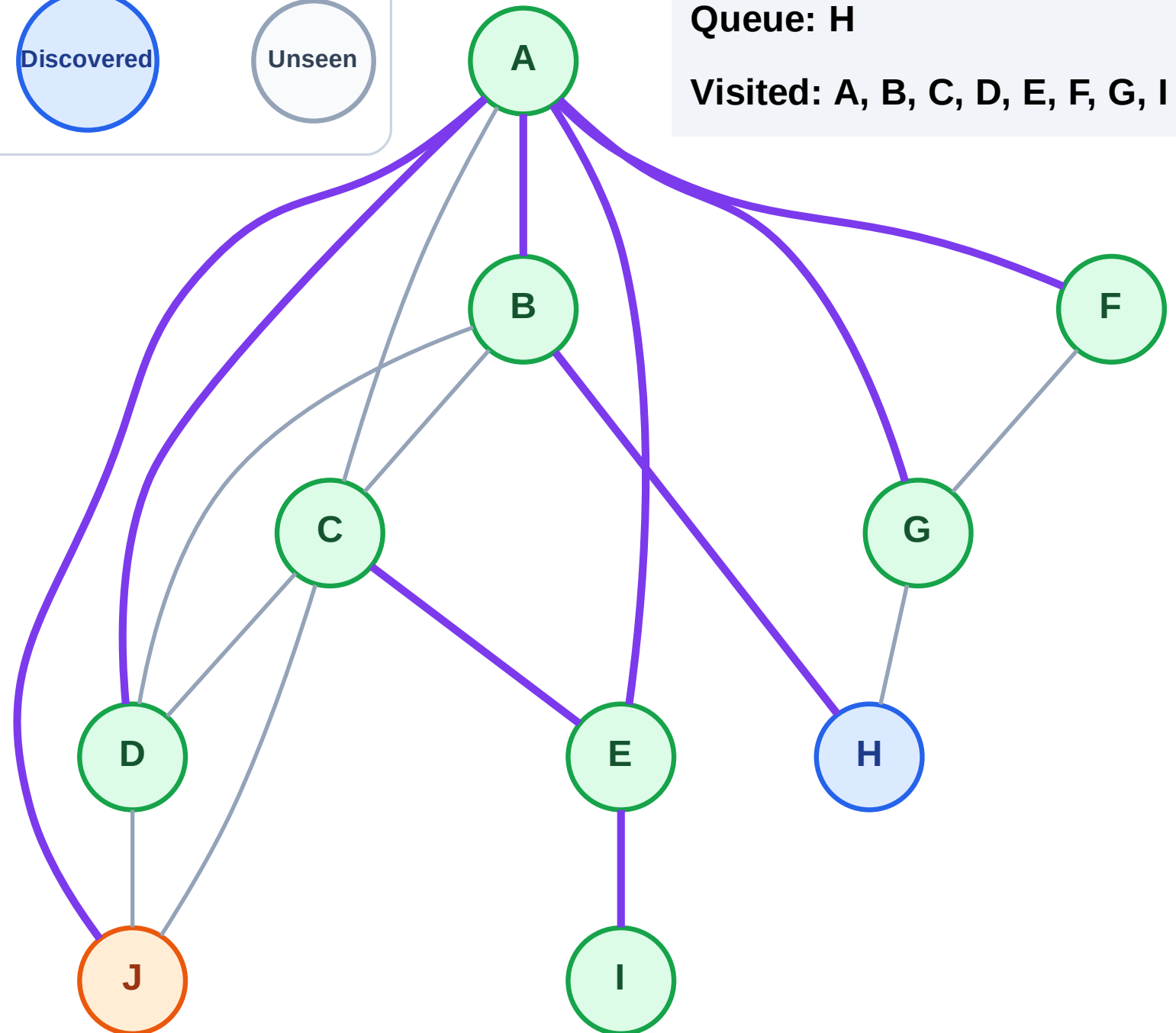
Processing

Discovered

Unseen

Queue: H

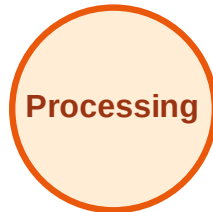
Visited: A, B, C, D, E, F, G, I



BFS Traversal

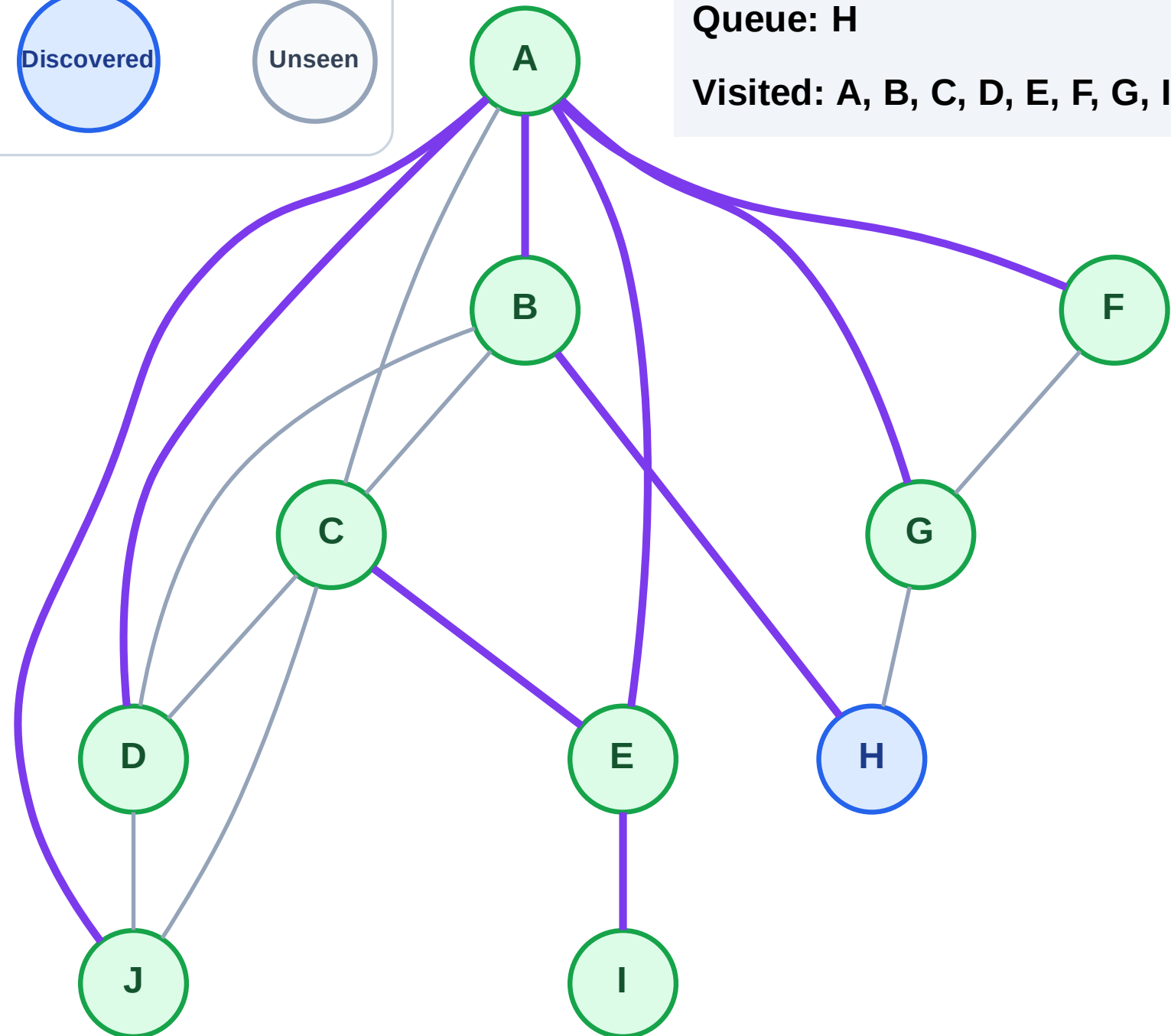
Step 19: No new neighbors from J

Legend



Queue: H

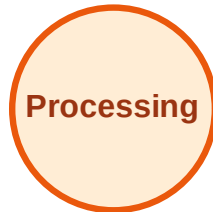
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

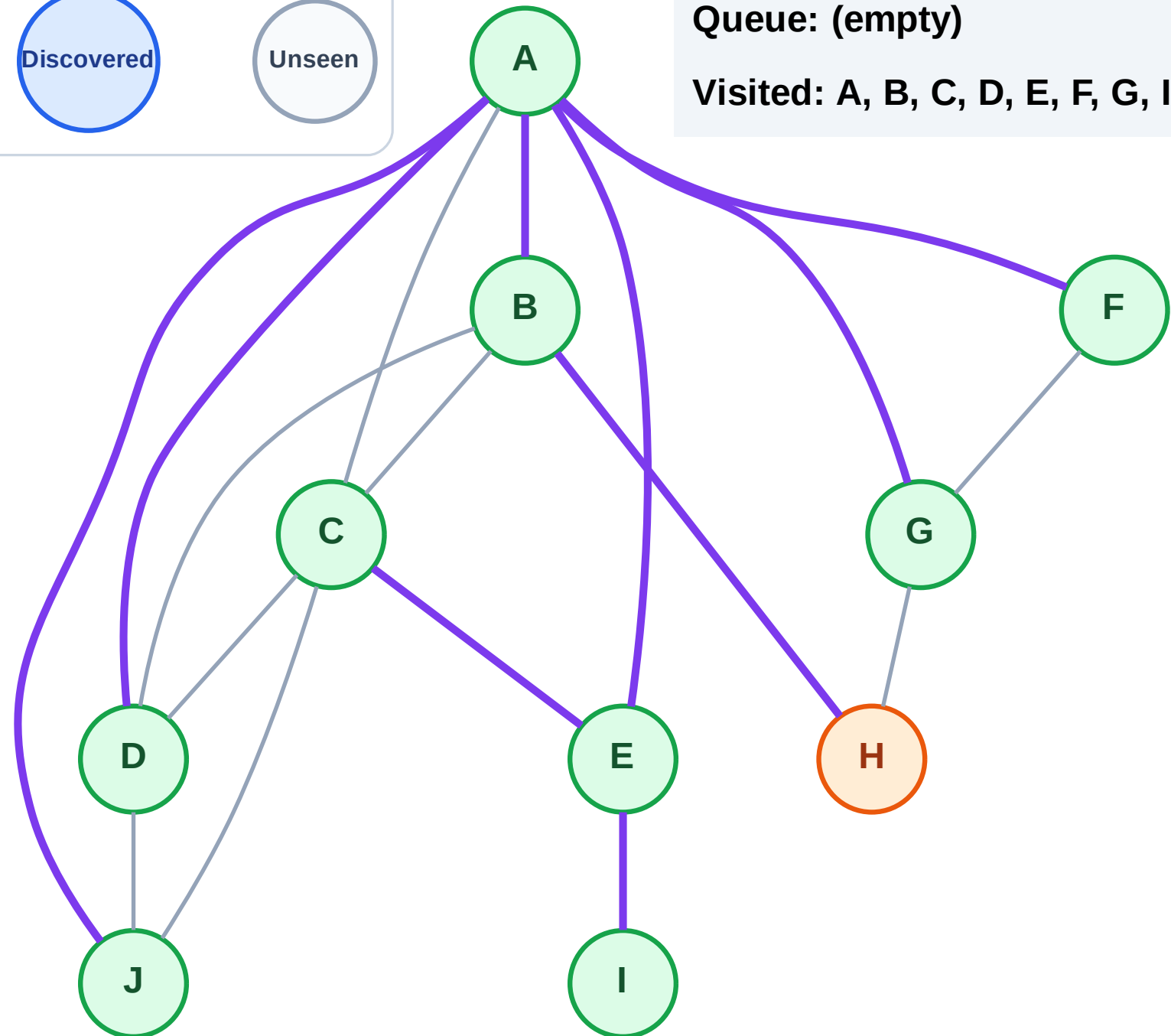
Step 20: Process node H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

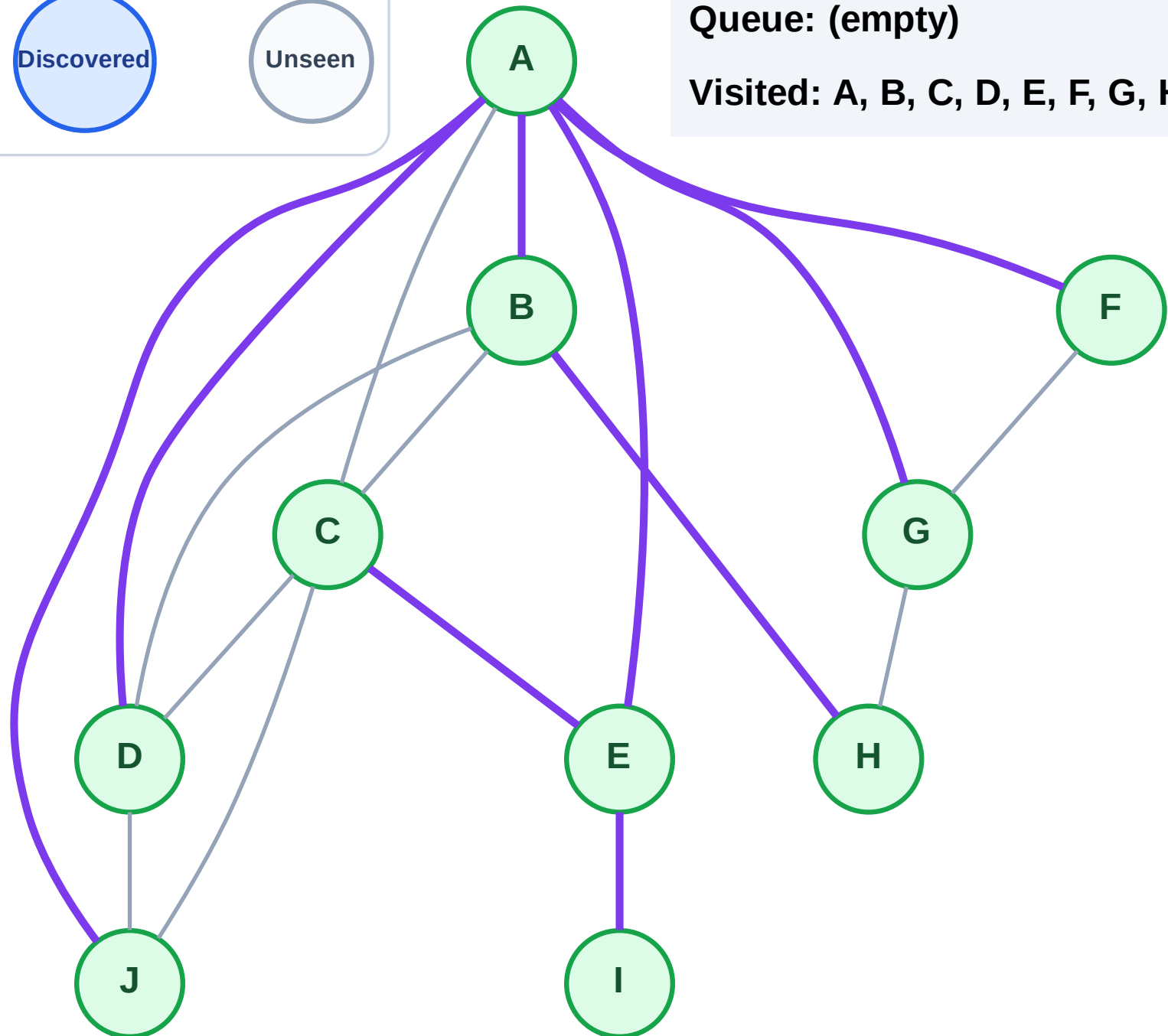
Step 21: No new neighbors from H

Legend



Queue: (empty)

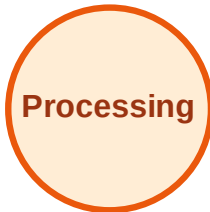
Visited: A, B, C, D, E, F, G, H, I, J



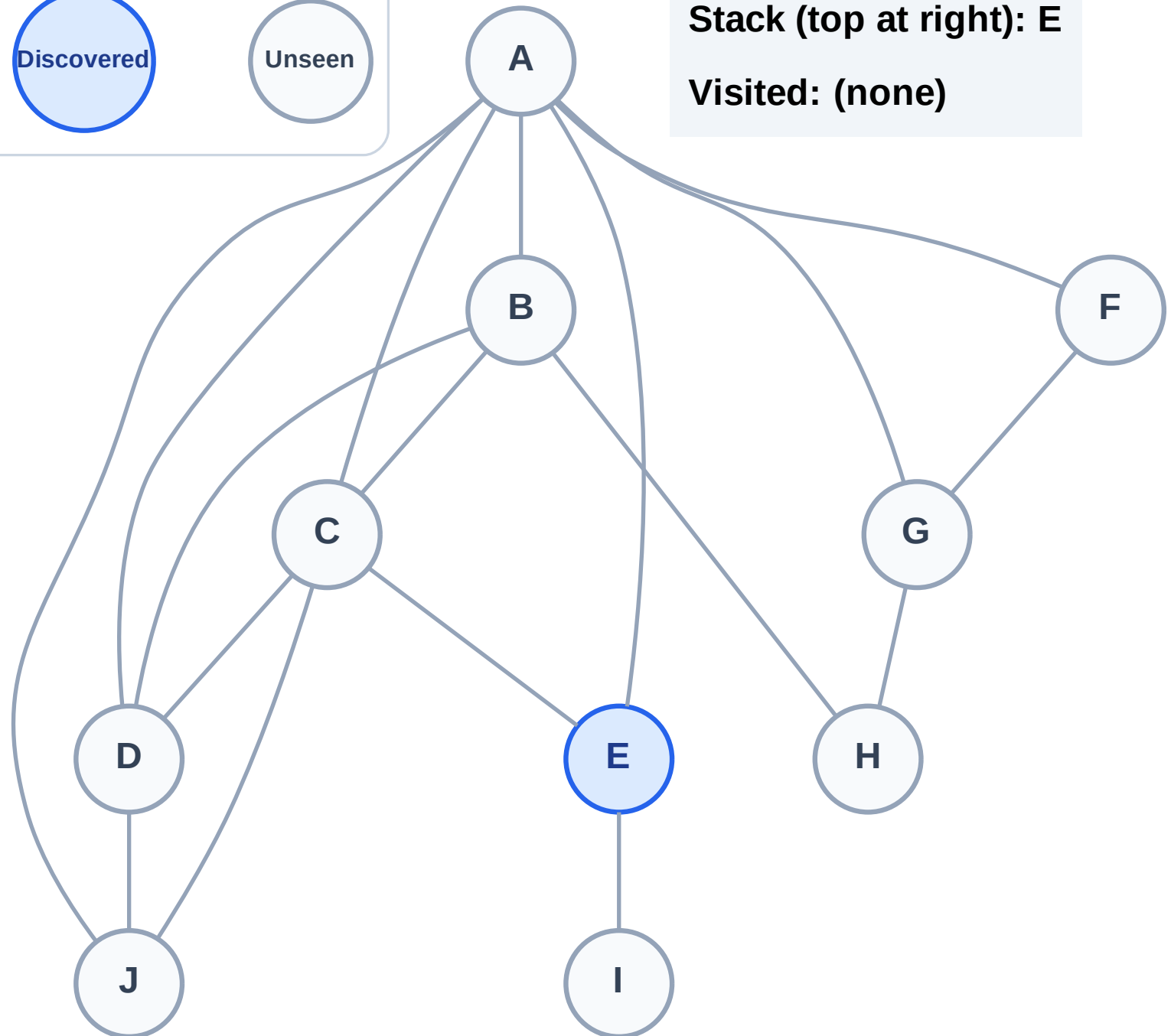
DFS Traversal

Step 1: Start at E

Legend



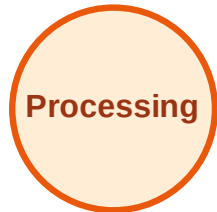
Stack (top at right): E
Visited: (none)



DFS Traversal

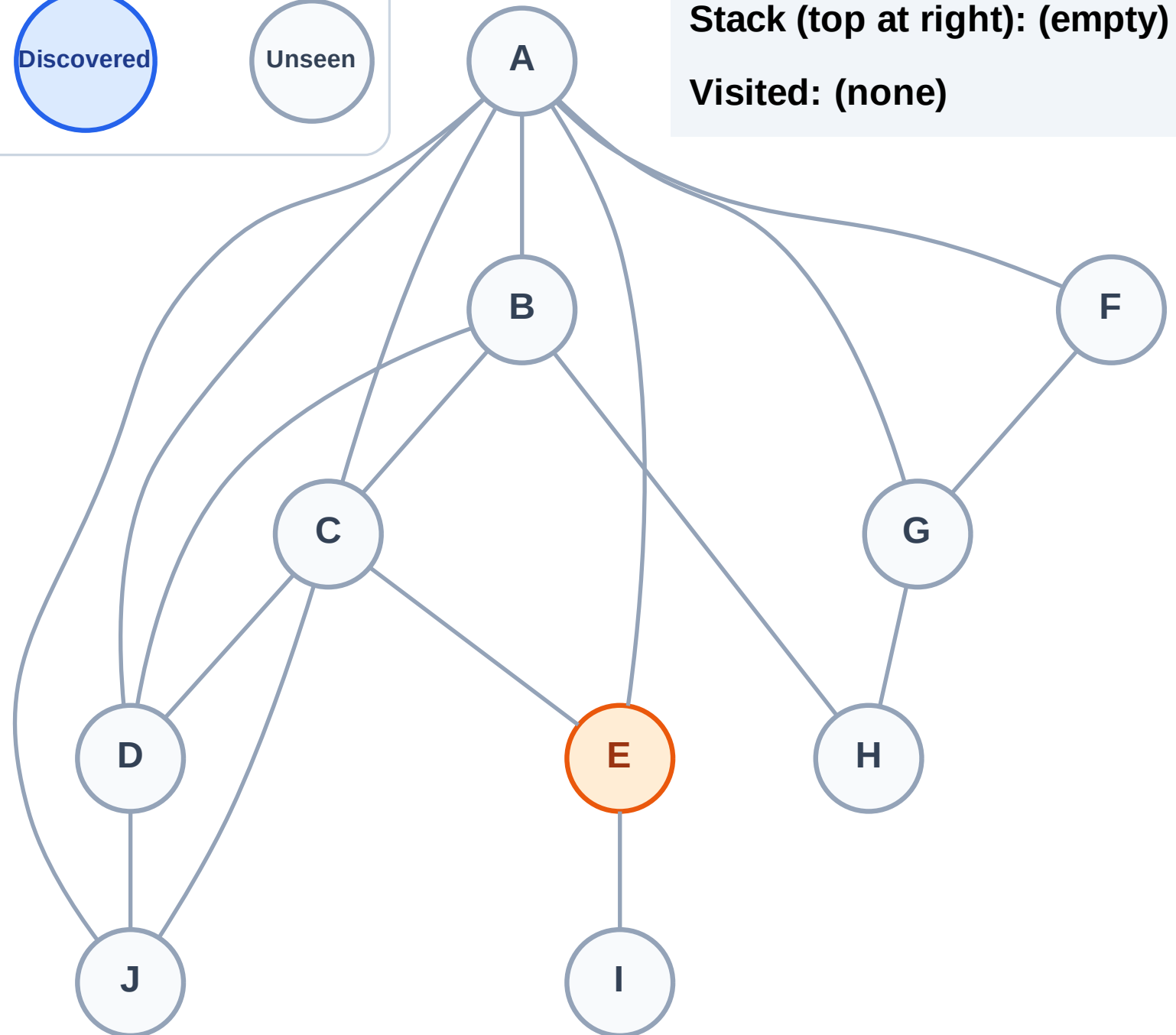
Step 2: Process node E

Legend



Stack (top at right): (empty)

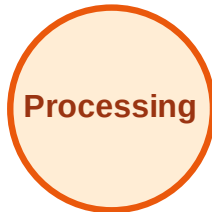
Visited: (none)



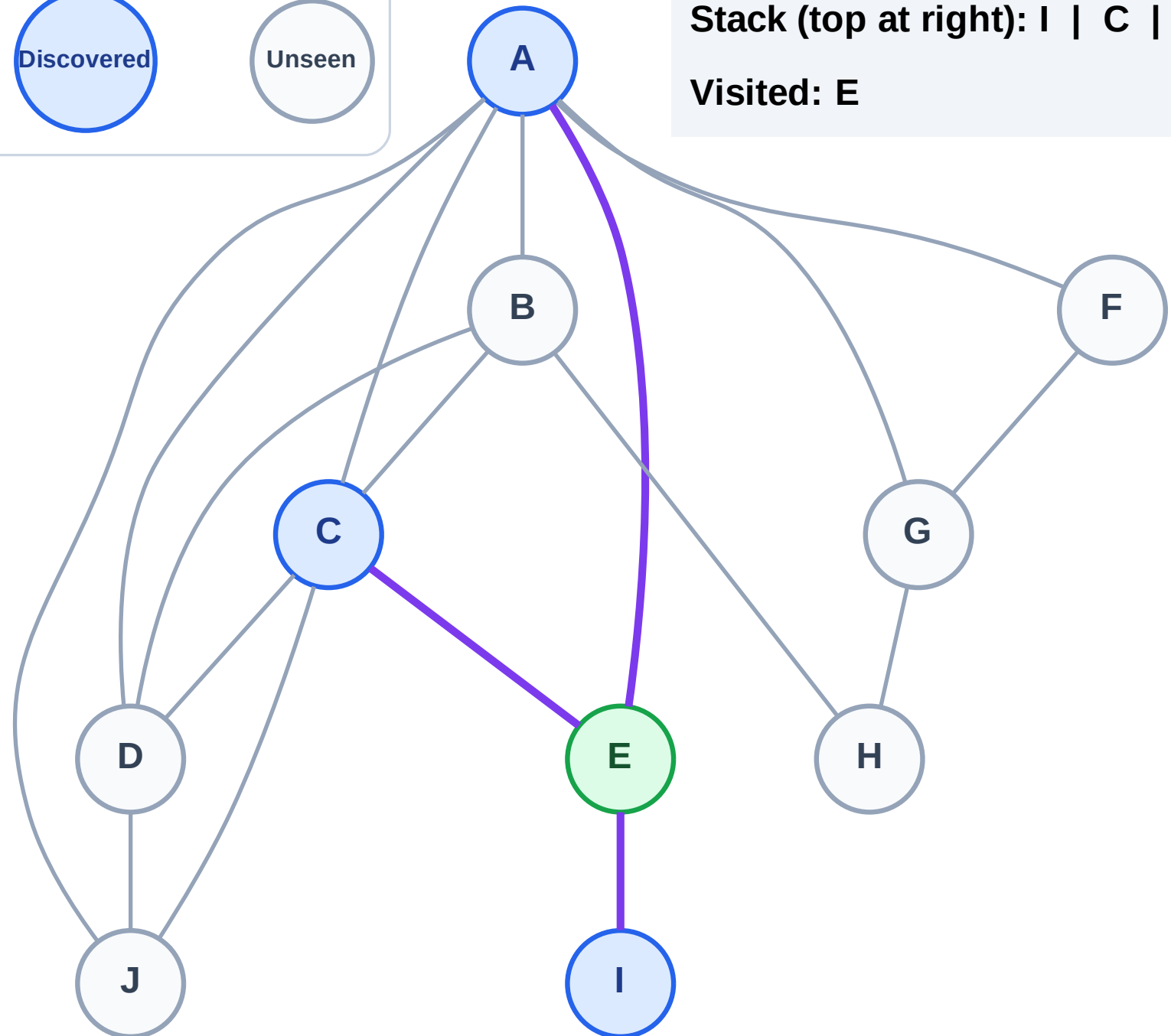
DFS Traversal

Step 3: Discover I, C, A from E

Legend



Stack (top at right): I | C | A
Visited: E



DFS Traversal

Step 4: Process node A

Legend

Complete

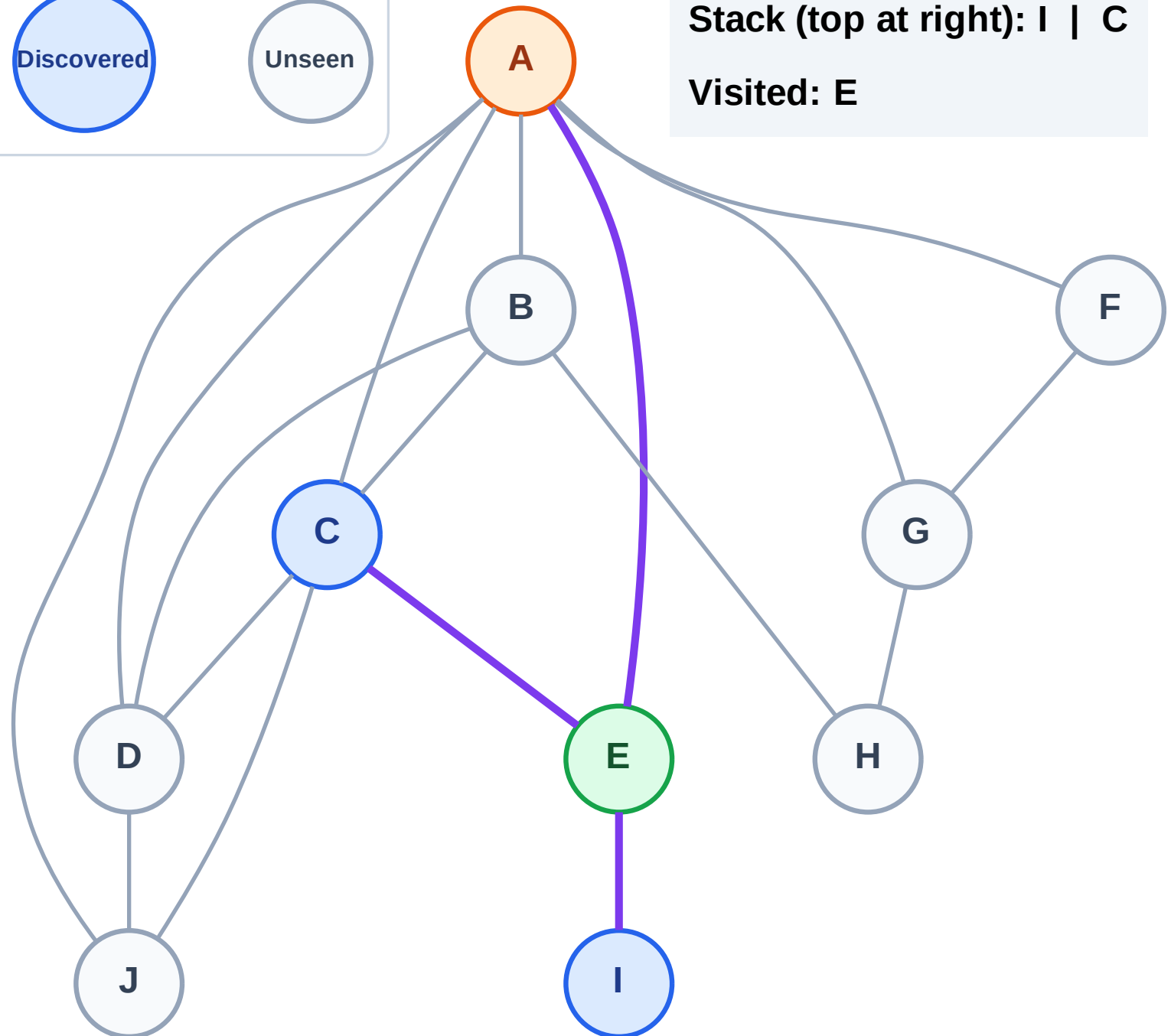
Processing

Discovered

Unseen

Stack (top at right): I | C

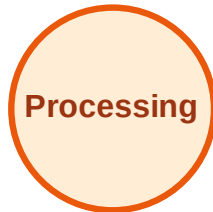
Visited: E



DFS Traversal

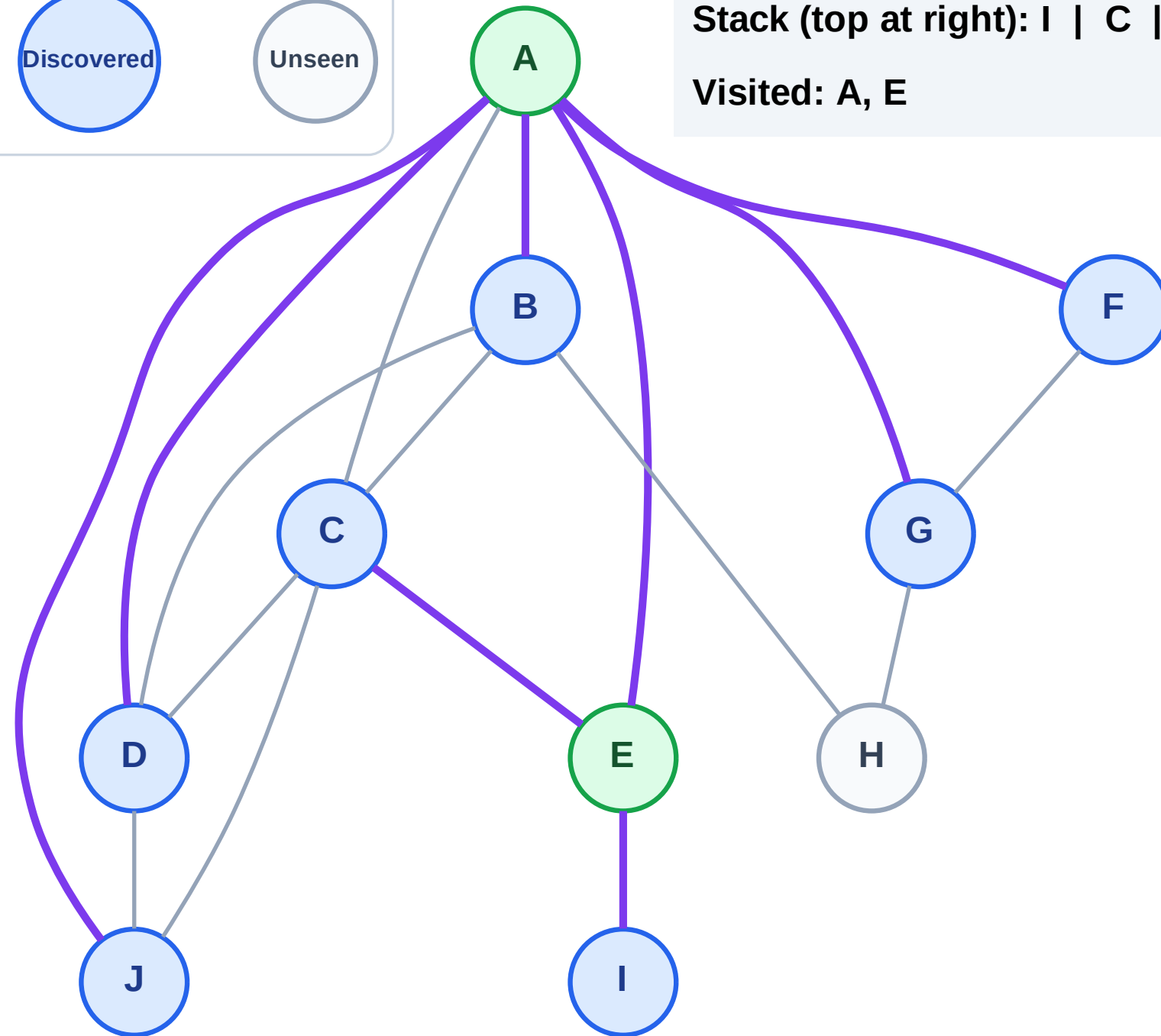
Step 5: Discover J, G, F, D, B from A

Legend



Stack (top at right): I | C | J | G | F | D | B

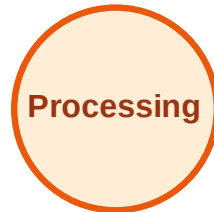
Visited: A, E



DFS Traversal

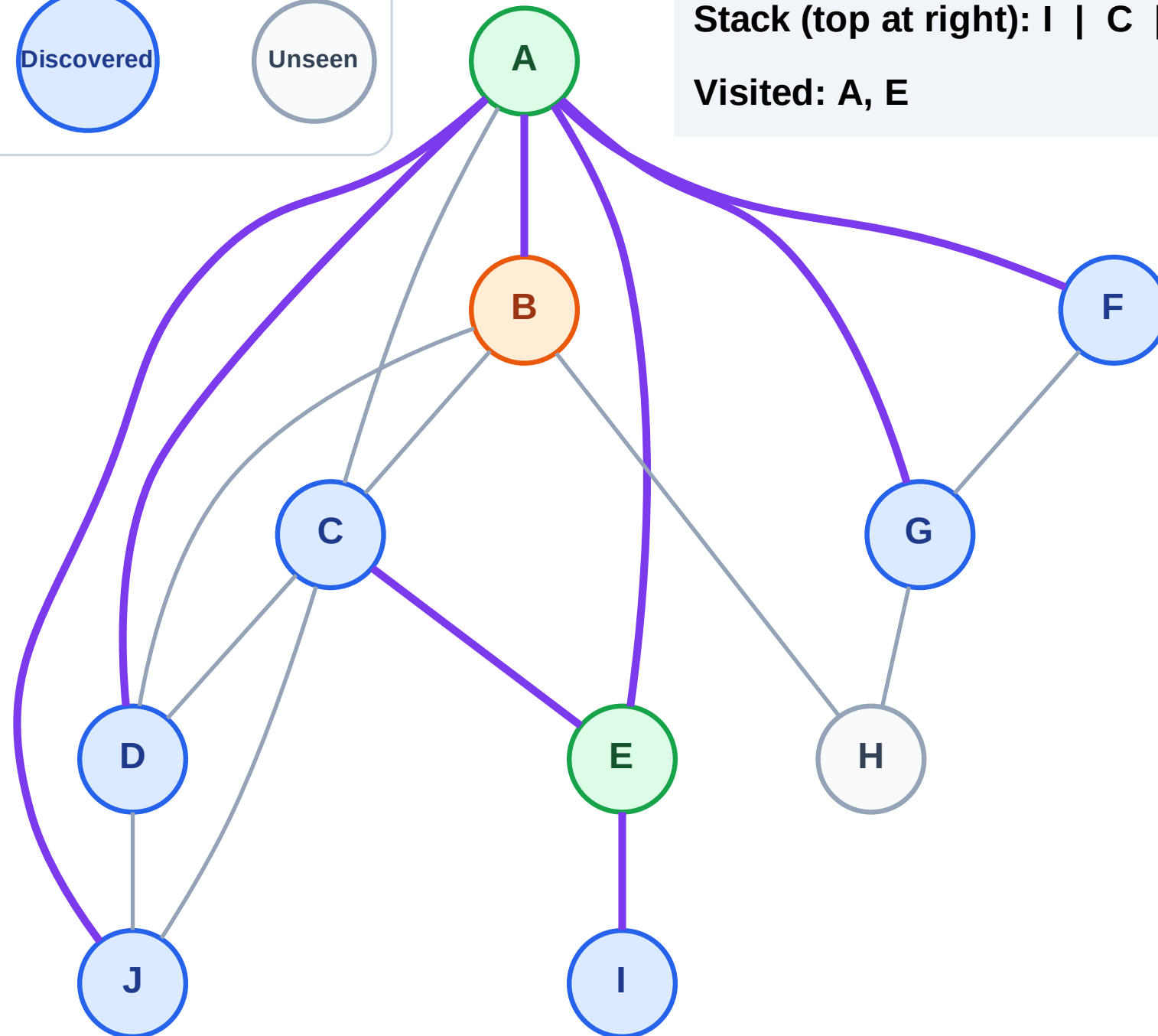
Step 6: Process node B

Legend



Stack (top at right): I | C | J | G | F | D

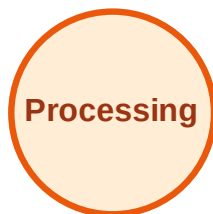
Visited: A, E



DFS Traversal

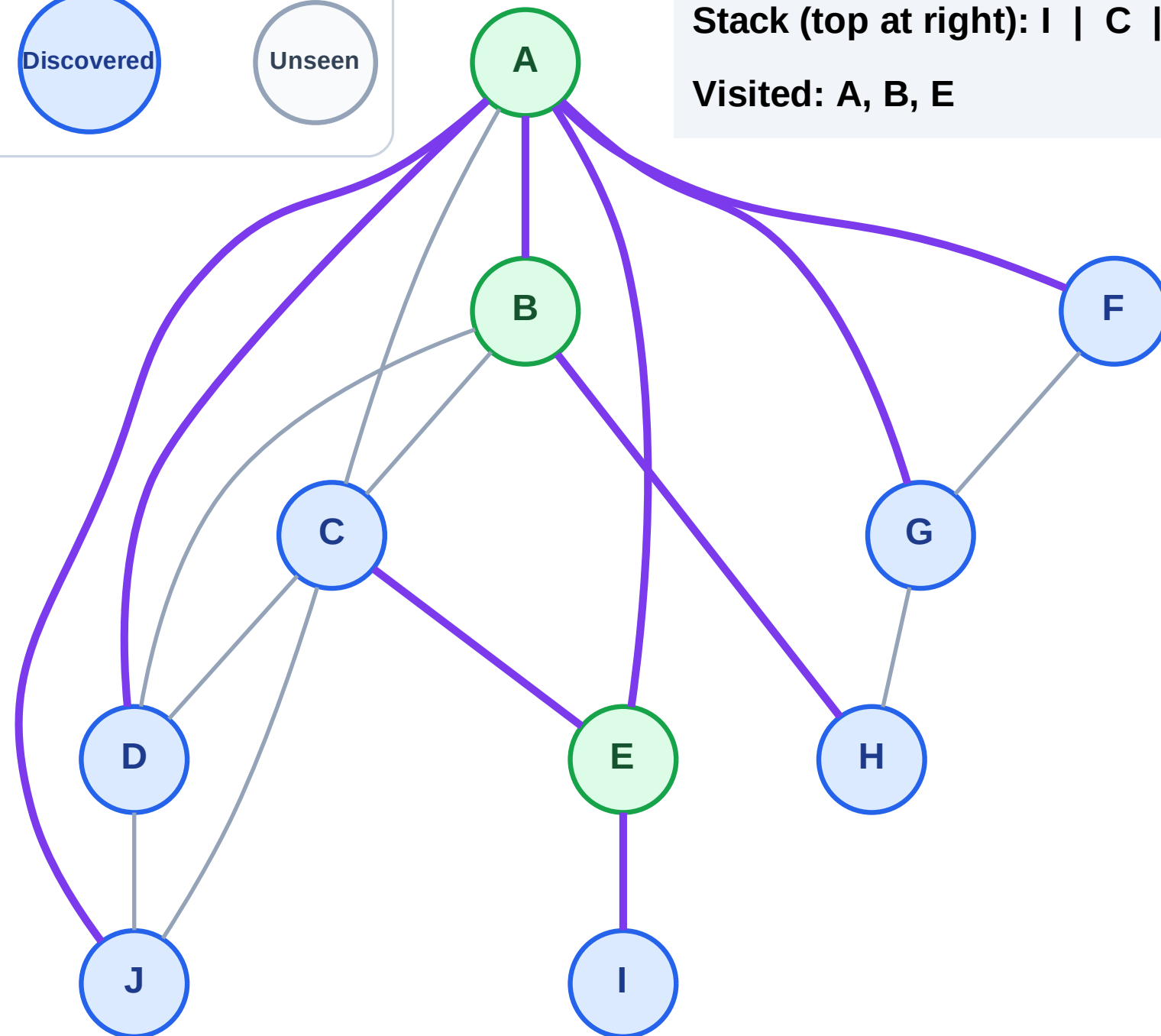
Step 7: Discover H from B

Legend

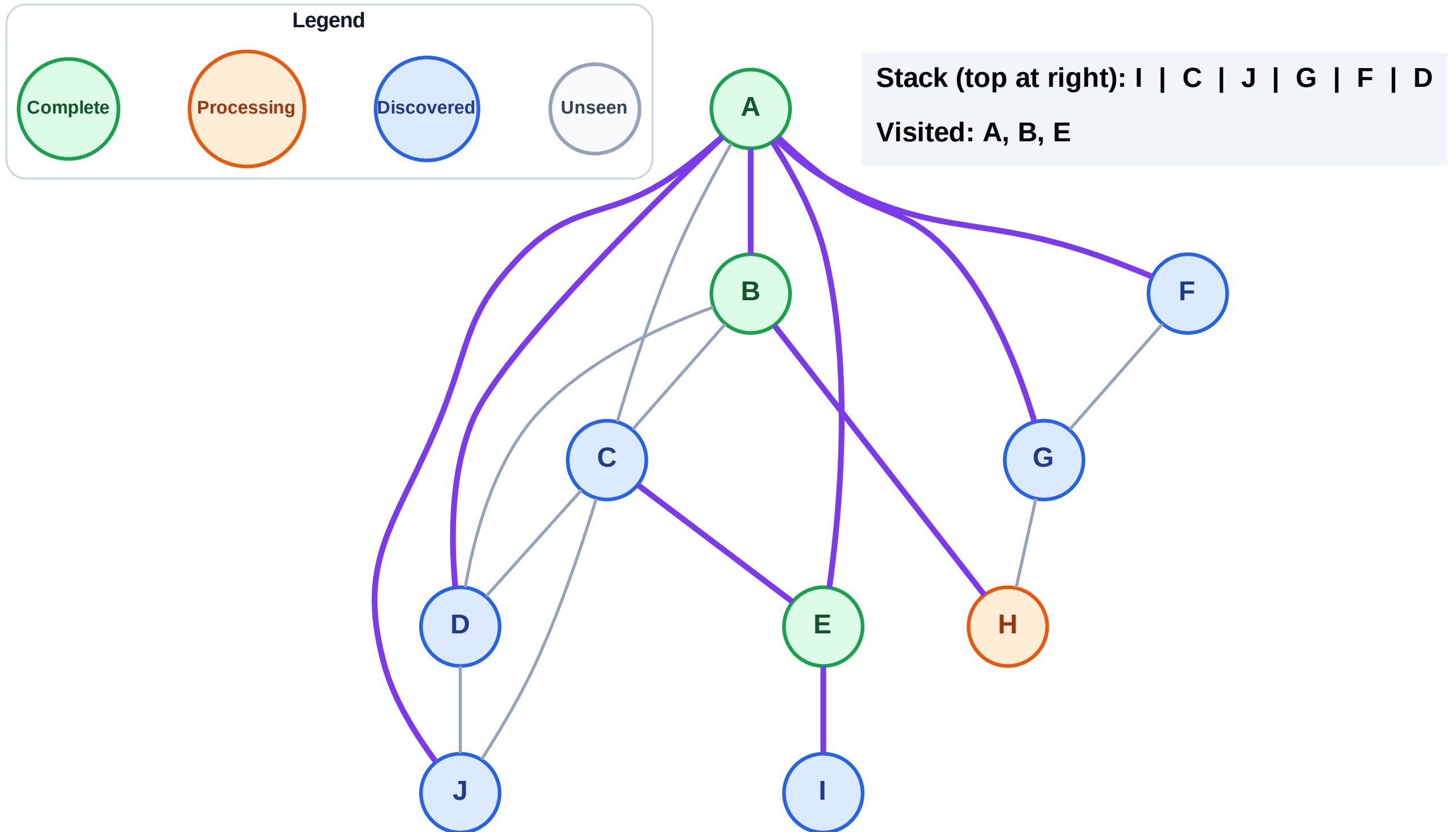


Stack (top at right): I | C | J | G | F | D | H

Visited: A, B, E

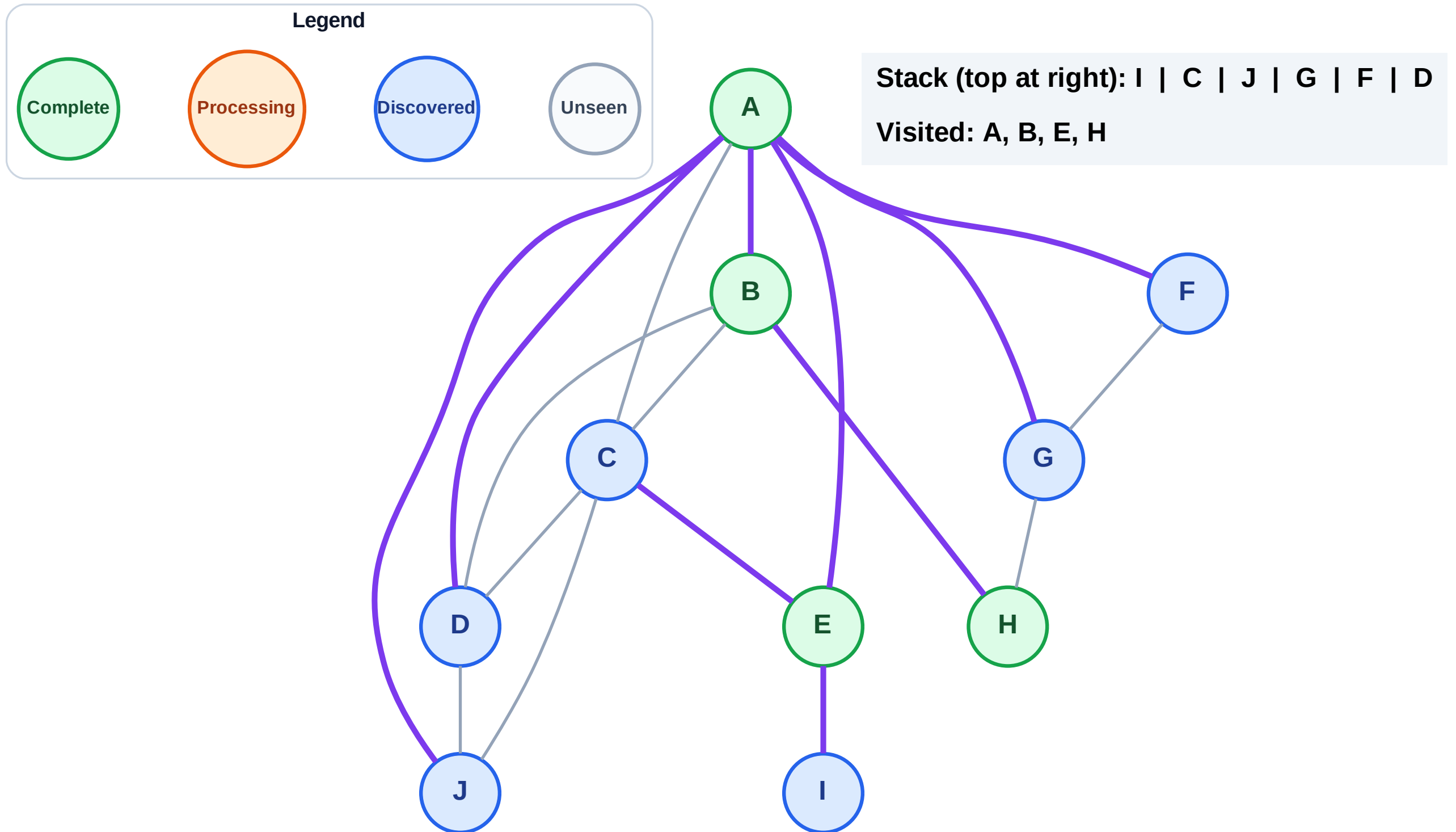


Step 8: Process node H



Step 9: No new neighbors from H

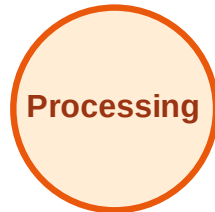
Step 9: No new neighbors from H



DFS Traversal

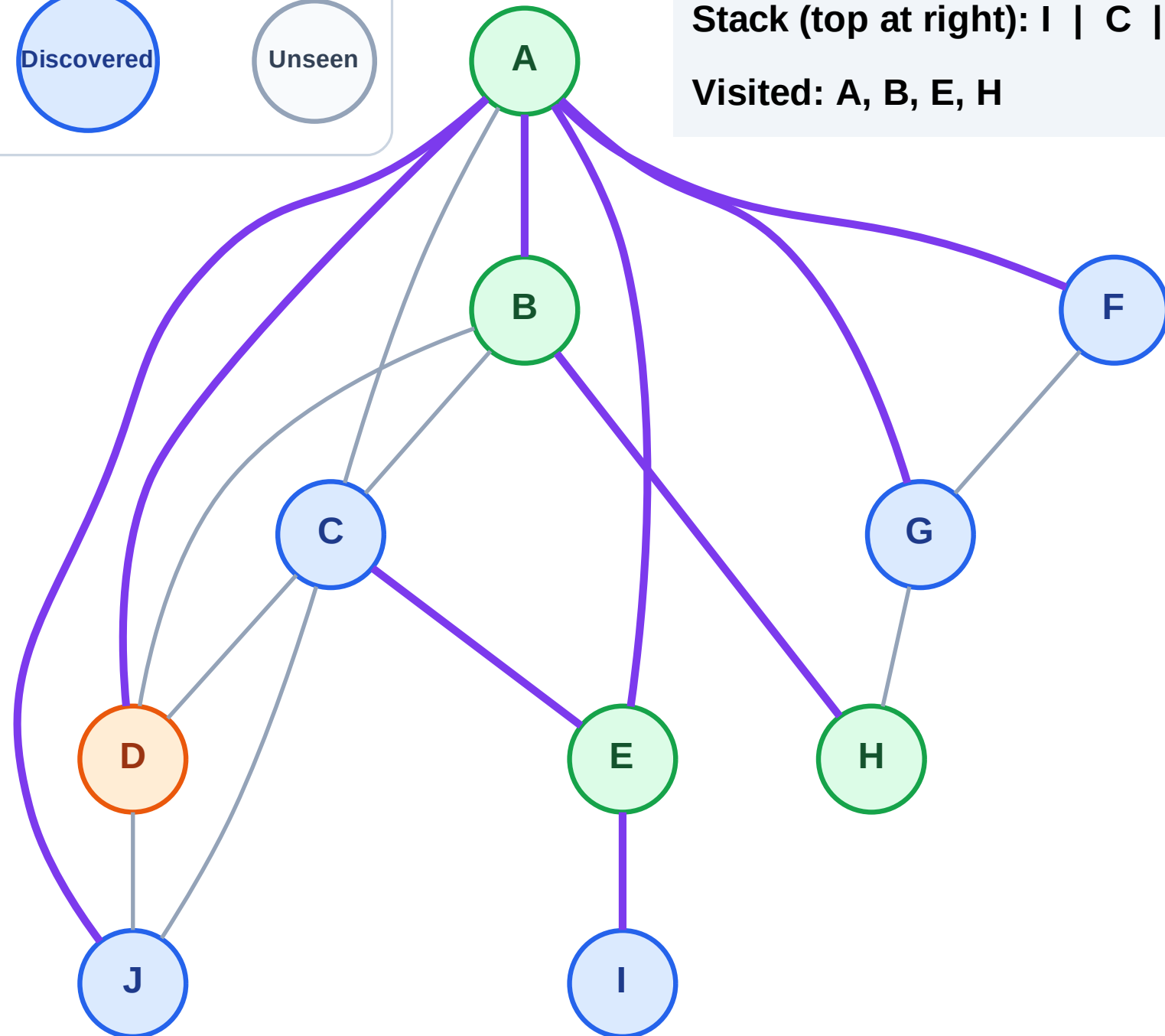
Step 10: Process node D

Legend



Stack (top at right): I | C | J | G | F

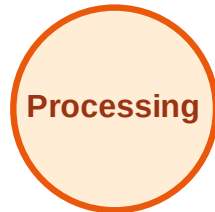
Visited: A, B, E, H



DFS Traversal

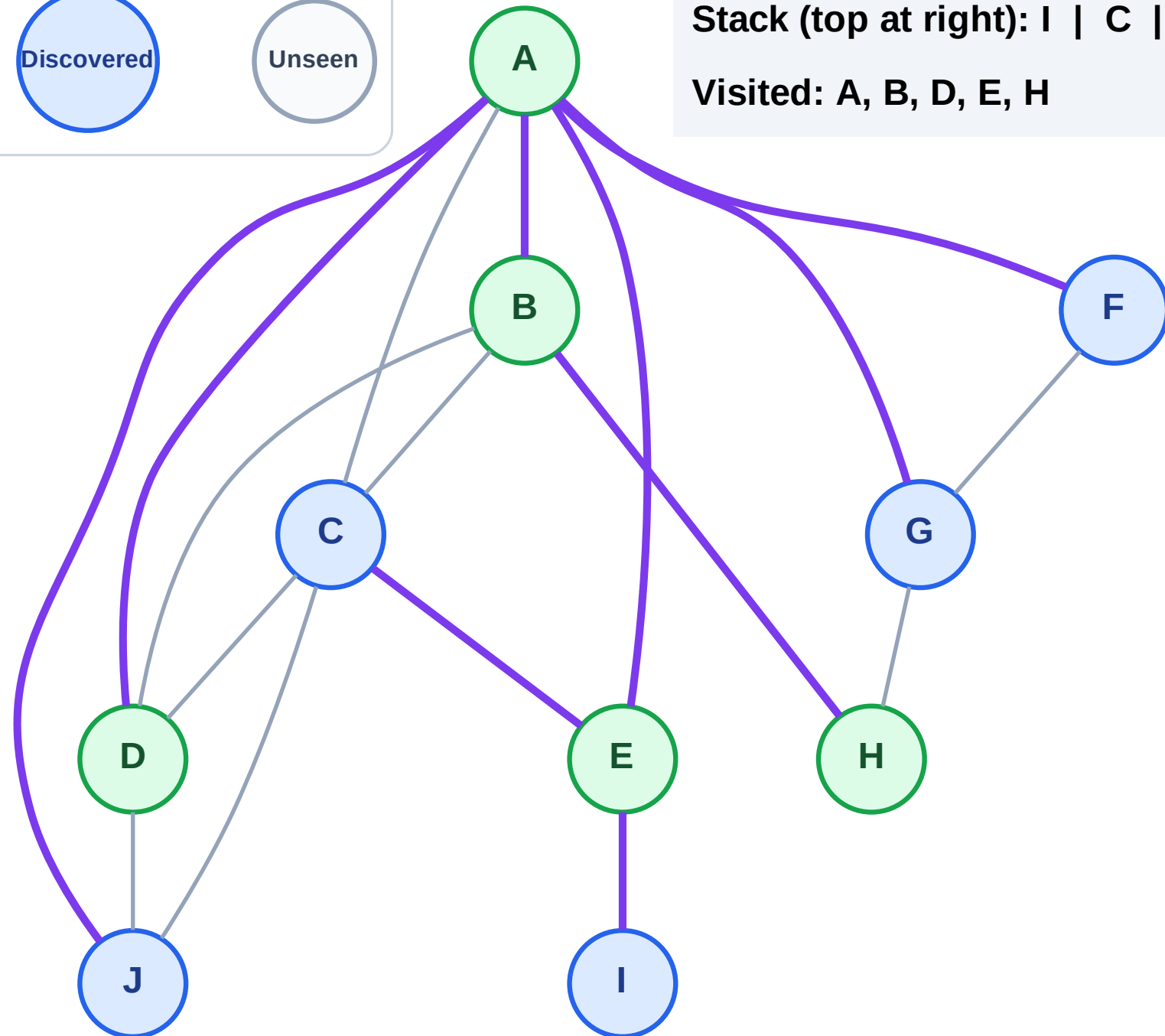
Step 11: No new neighbors from D

Legend



Stack (top at right): I | C | J | G | F

Visited: A, B, D, E, H



DFS Traversal

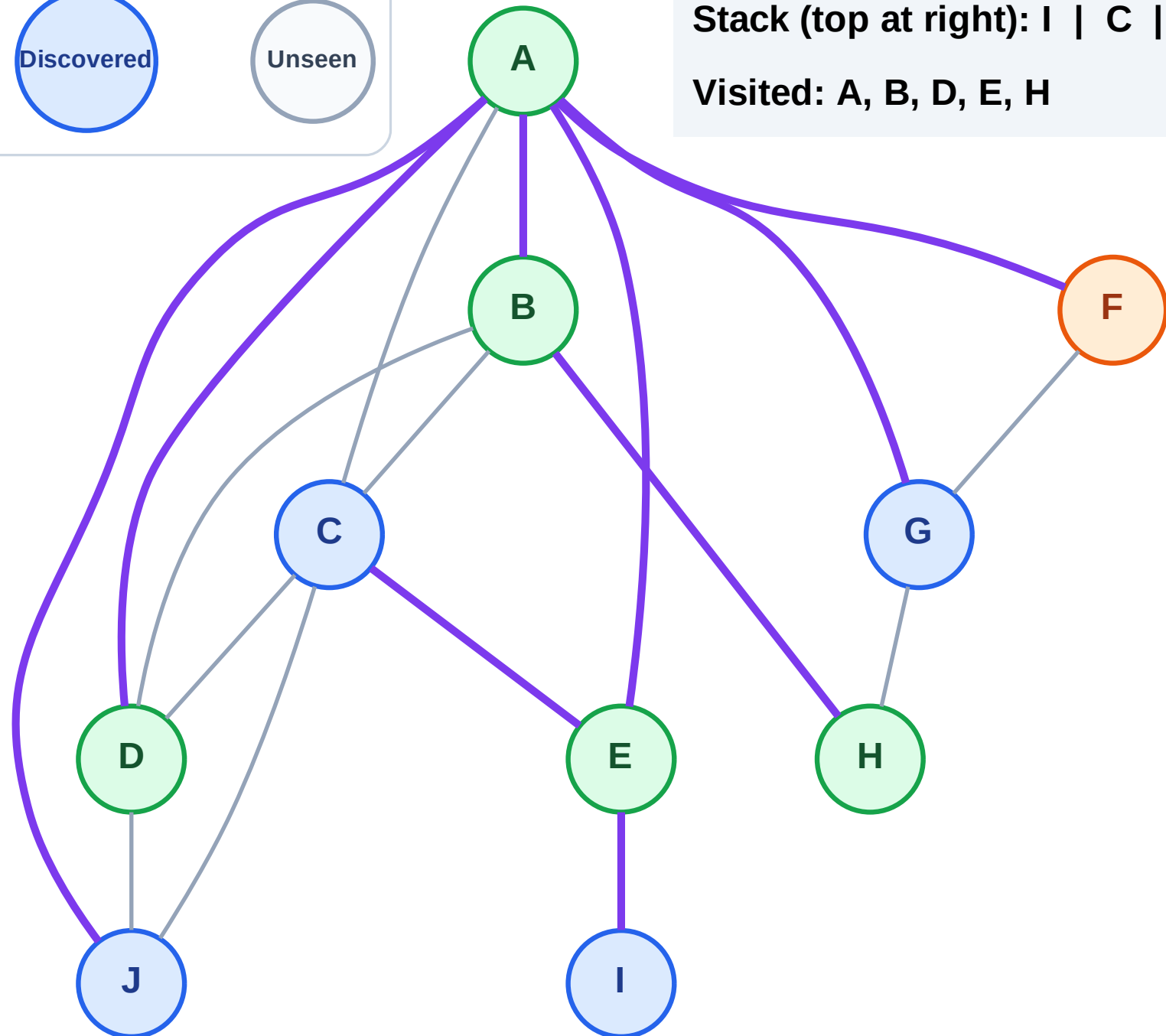
Step 12: Process node F

Legend



Stack (top at right): I | C | J | G

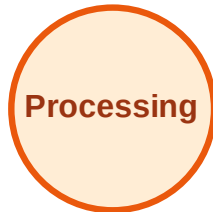
Visited: A, B, D, E, H



DFS Traversal

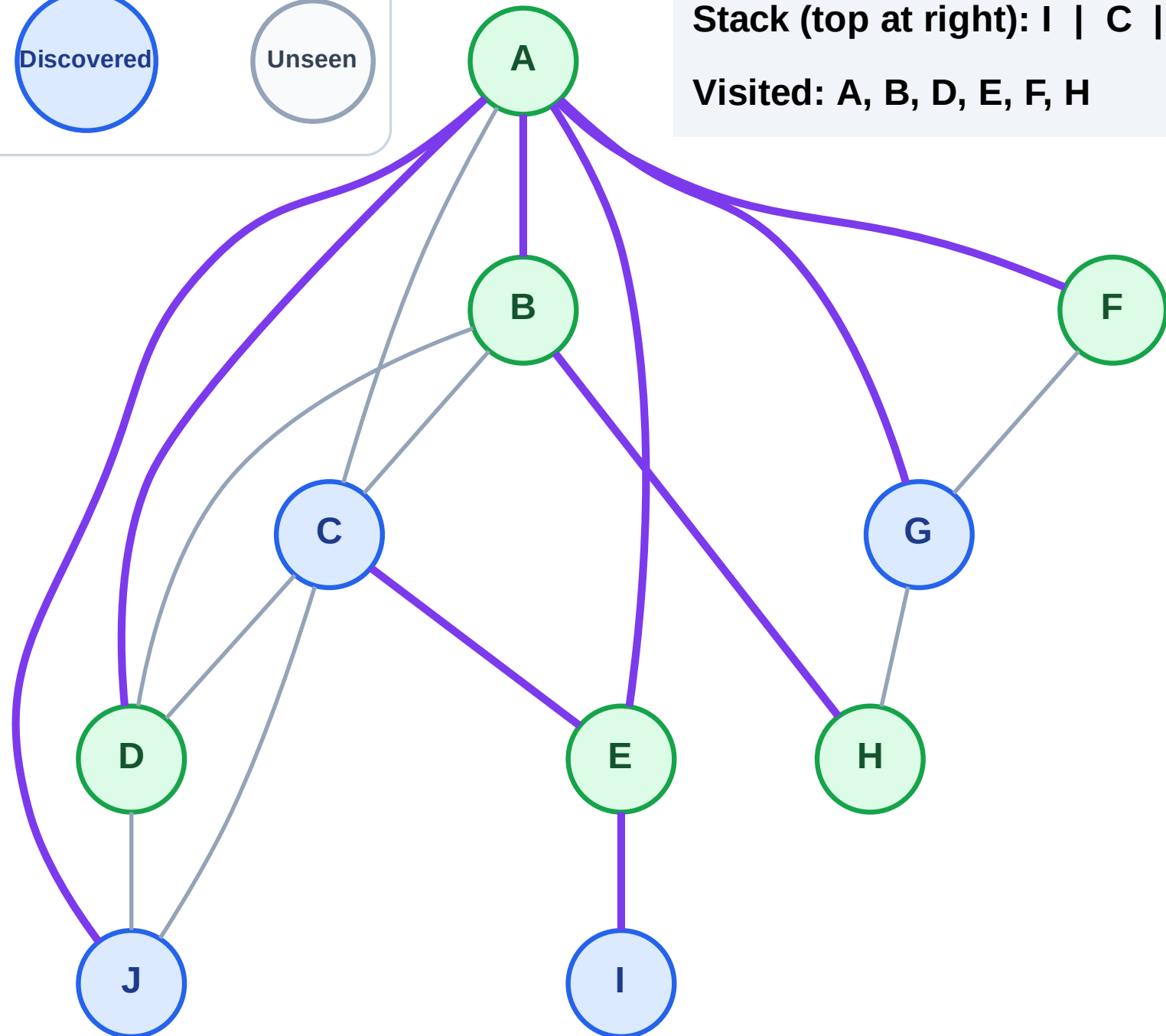
Step 13: No new neighbors from F

Legend



Stack (top at right): I | C | J | G

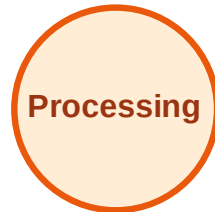
Visited: A, B, D, E, F, H



DFS Traversal

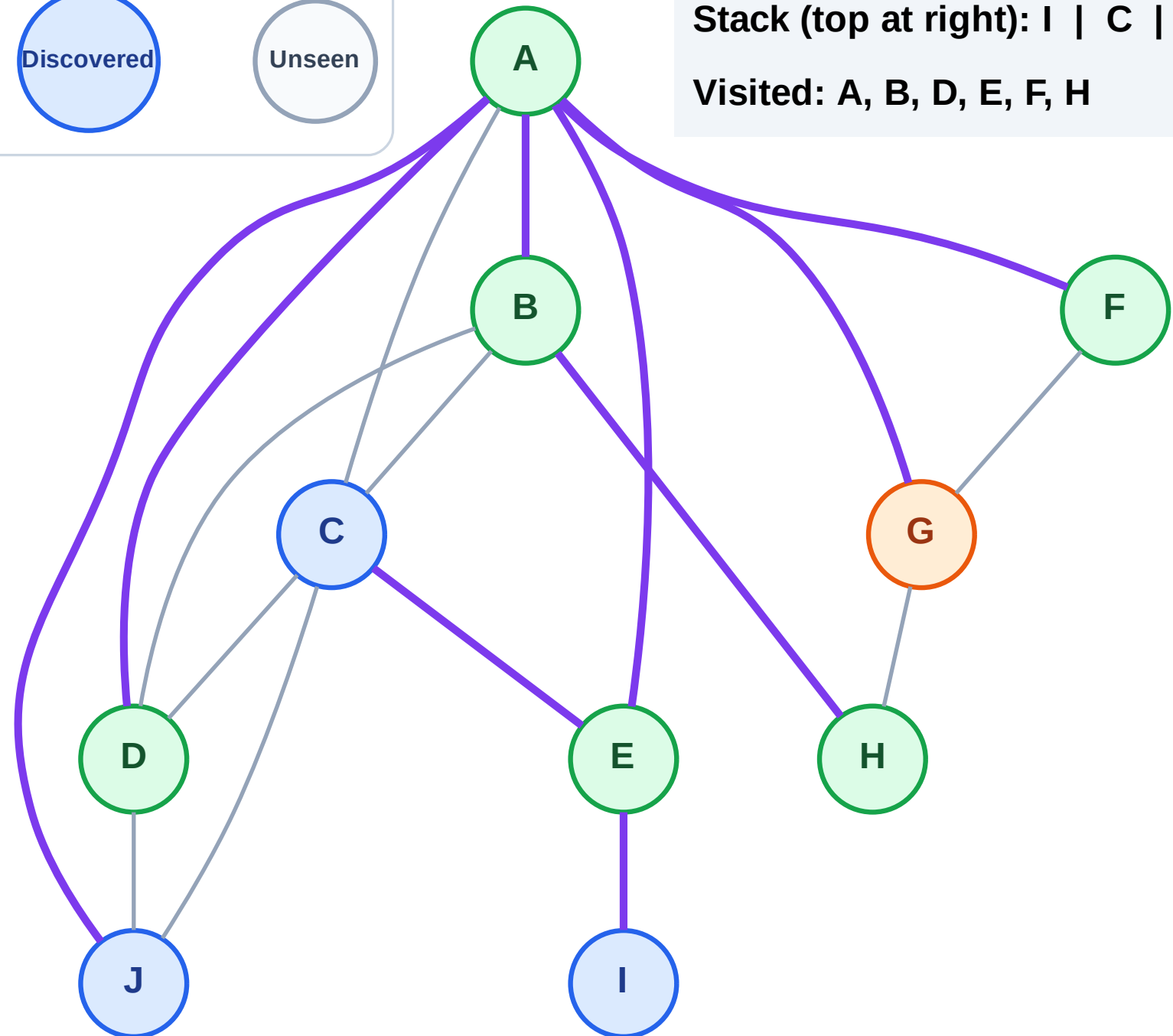
Step 14: Process node G

Legend



Stack (top at right): I | C | J

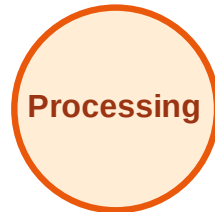
Visited: A, B, D, E, F, H



DFS Traversal

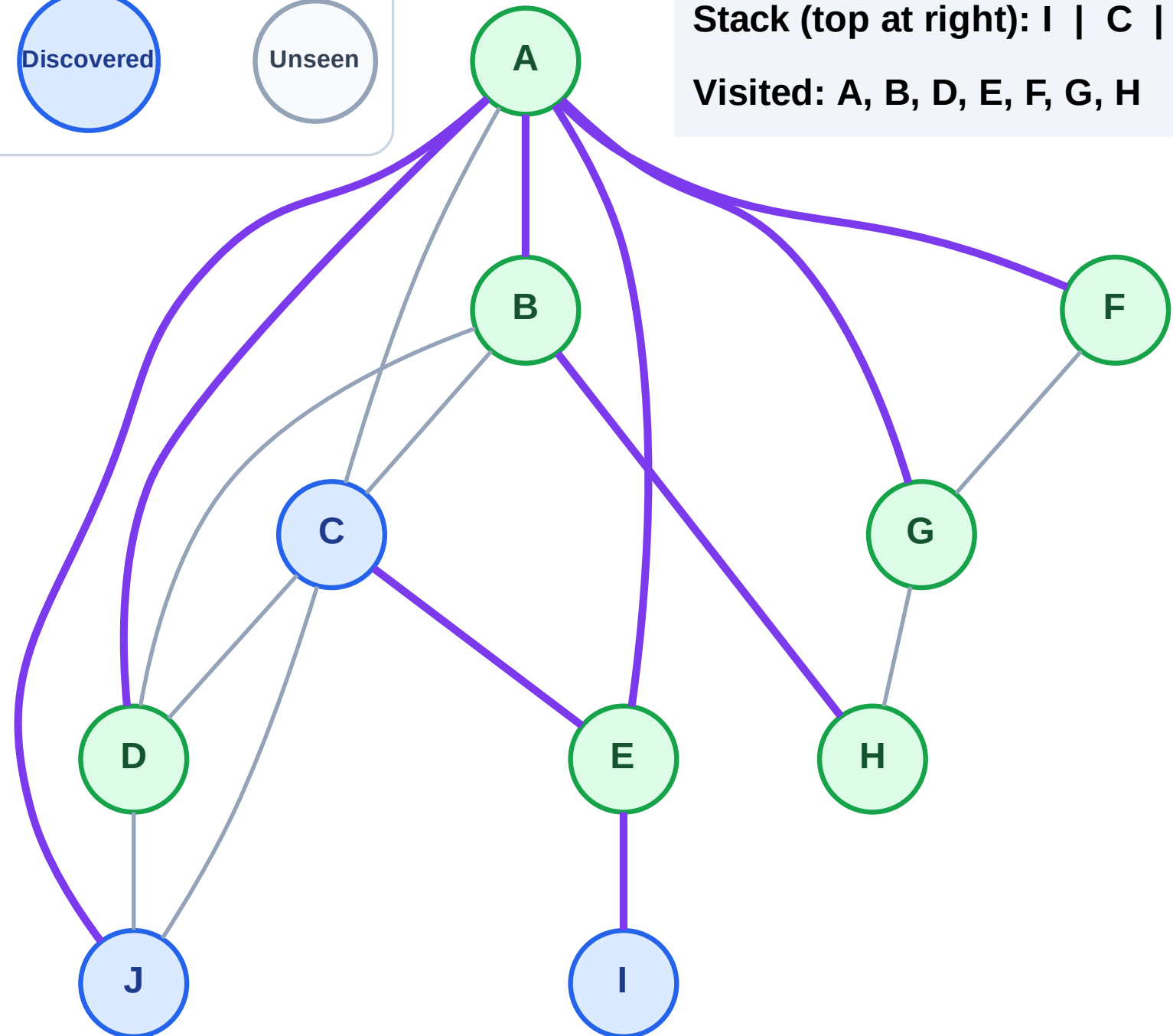
Step 15: No new neighbors from G

Legend



Stack (top at right): I | C | J

Visited: A, B, D, E, F, G, H



DFS Traversal

Step 16: Process node J

Legend

Complete

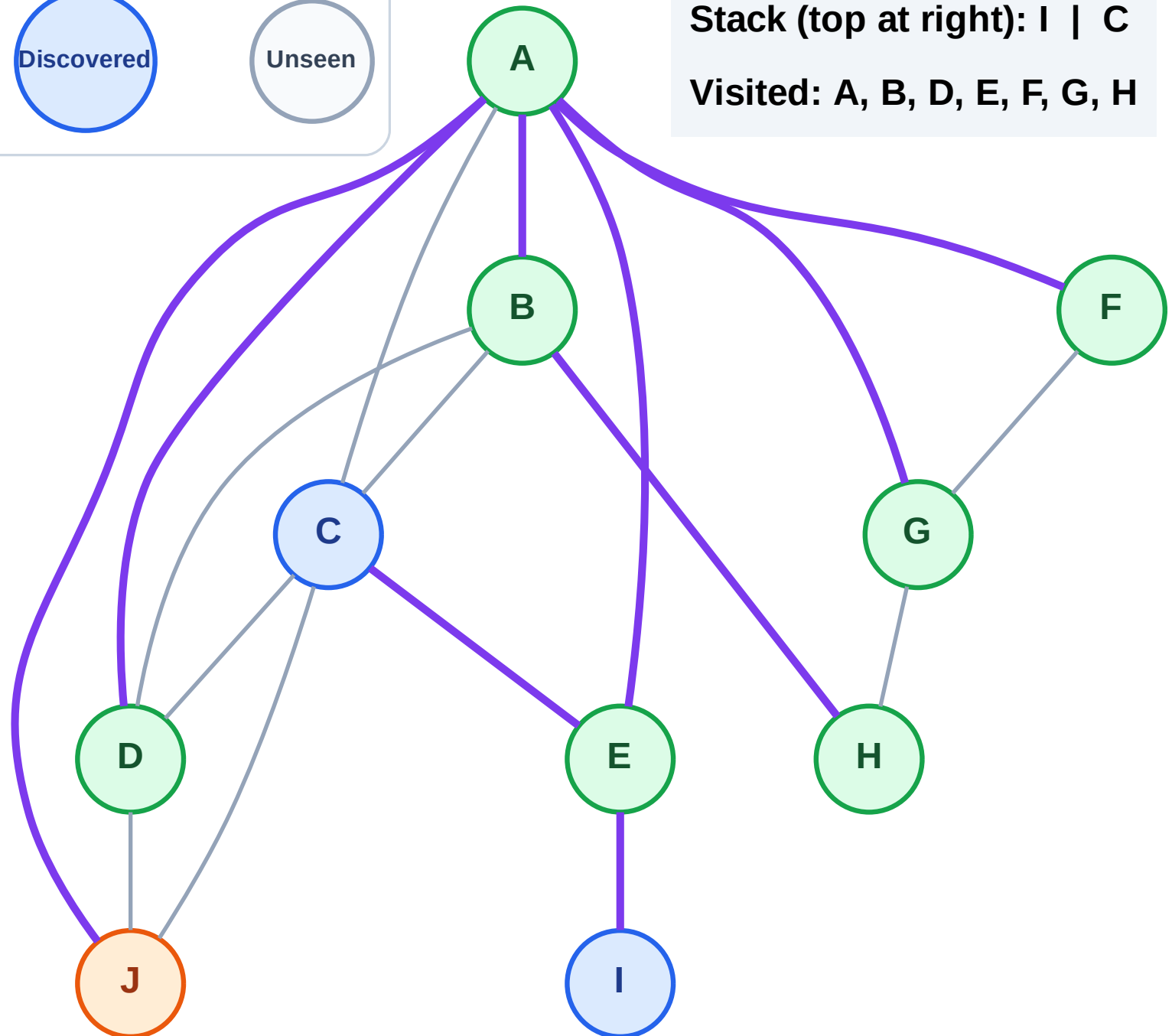
Processing

Discovered

Unseen

Stack (top at right): I | C

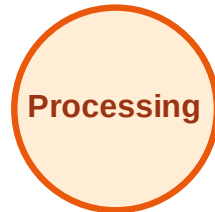
Visited: A, B, D, E, F, G, H



DFS Traversal

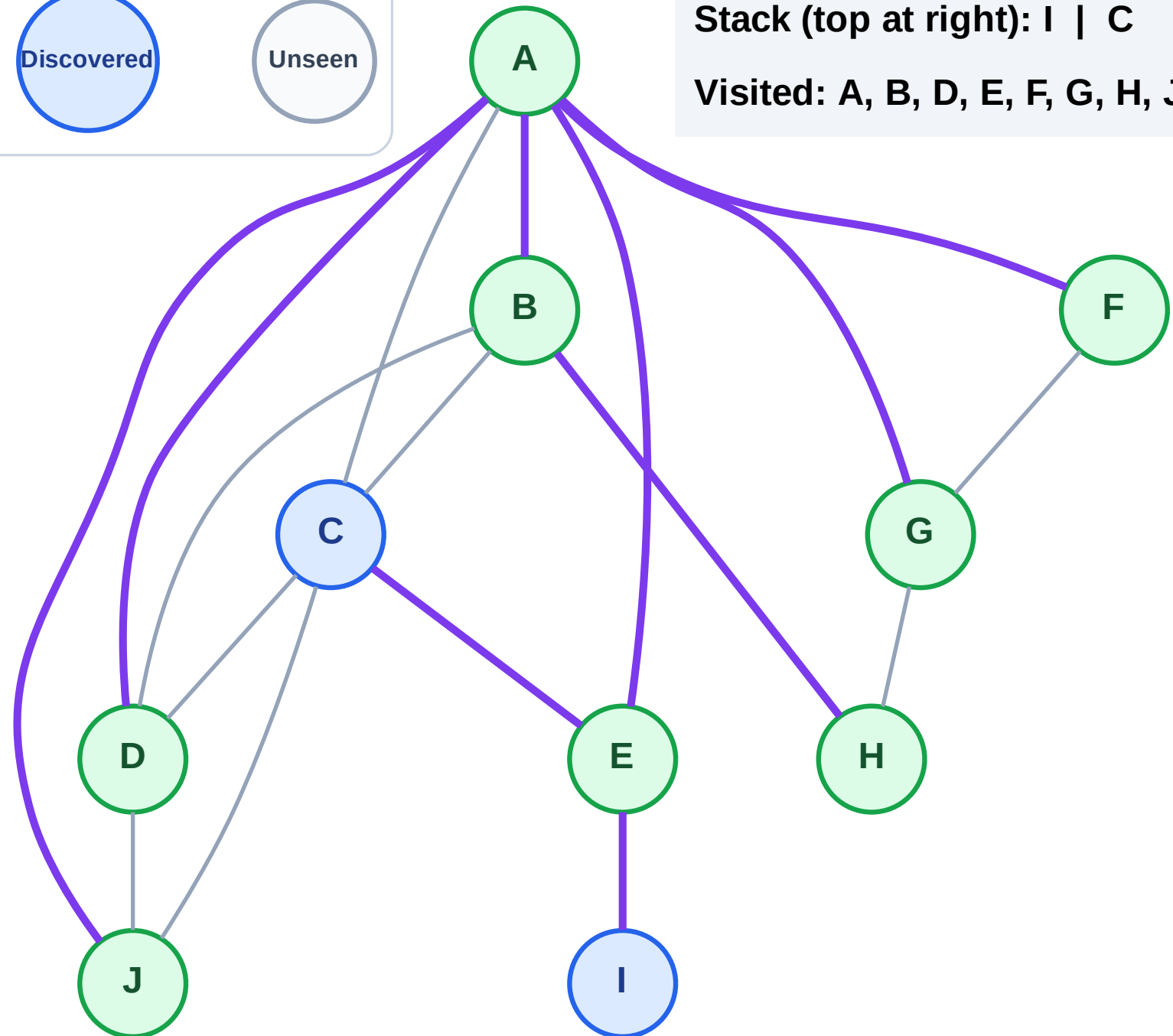
Step 17: No new neighbors from J

Legend



Stack (top at right): I | C

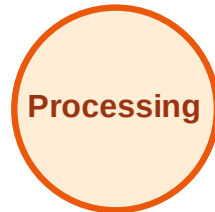
Visited: A, B, D, E, F, G, H, J



DFS Traversal

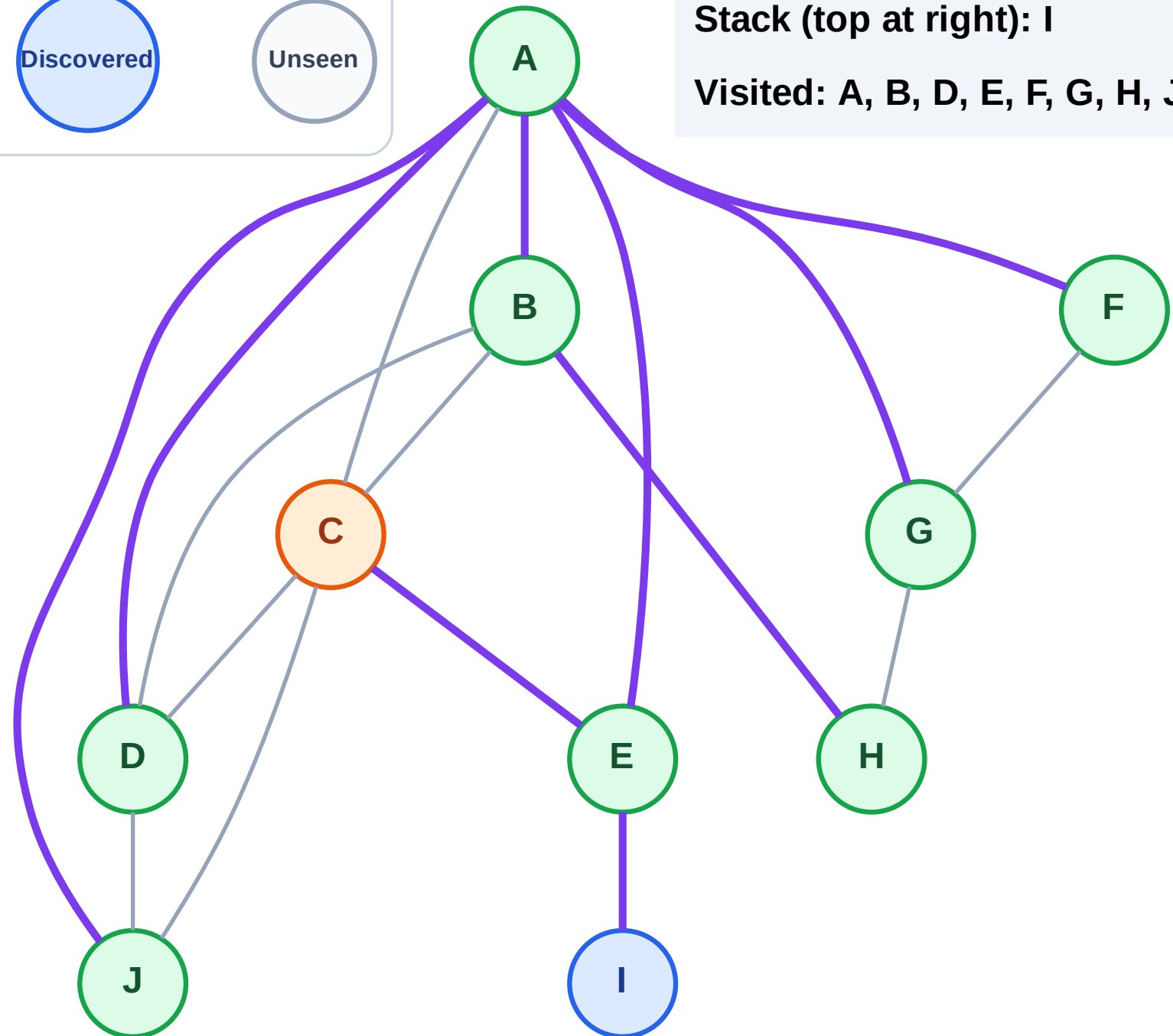
Step 18: Process node C

Legend



Stack (top at right): I

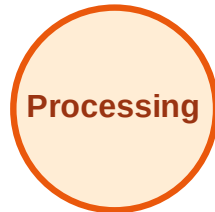
Visited: A, B, D, E, F, G, H, J



DFS Traversal

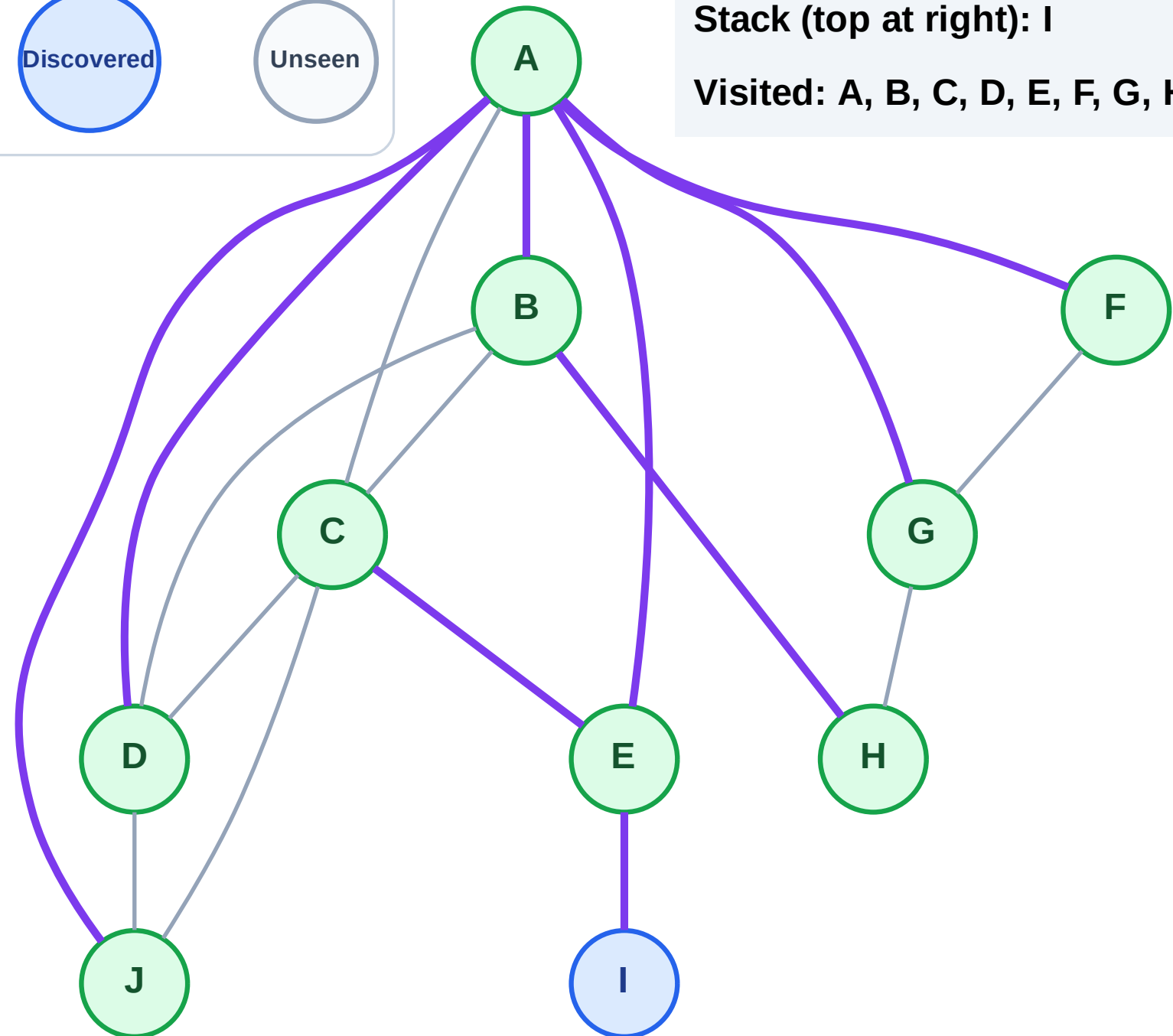
Step 19: No new neighbors from C

Legend



Stack (top at right): I

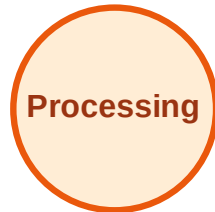
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

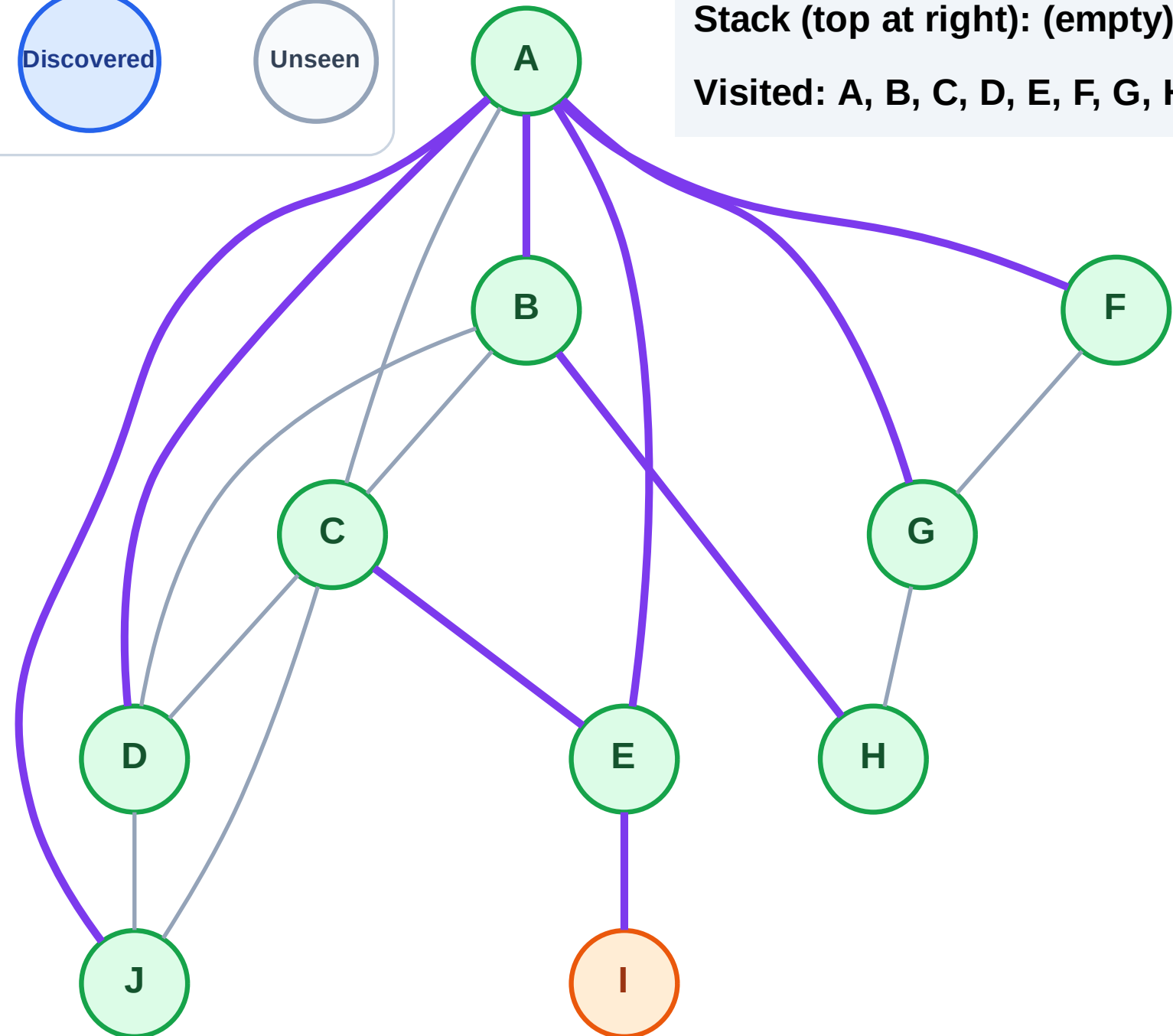
Step 20: Process node I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

